Malware Analysis Report For MachO-OSX-ppc-and-i386-bash

Binary Type

Mach-O (Mach Object). (Approved)

Analyzing File

- **File Name**: MachO-OSX-ppc-and-i386-bash
- **File Extension**:
- **File Type**: Mach-O (Mach Object).
- **MD5 Hash**: 87d06d2032ce394b4c0a0c9f2c312174
- **SHA256 Hash**: 5c90ec3507016430a6067da37e1fcfc9faebd663b62032bc46524af667191ddd

File Identification

File detected as a valid macOS/iOS executable.

Mach-O File Analysis

Mach-O File Analysis: True

Mach-O File Imports

No imported functions were found.

Mach-O File Exports

No exported functions were found.

String Analysis

ASCII Strings

- __PAGEZERO
- __TEXT
- __text
- __TEXT
- __cstring
- __TEXT
- __const
- __TEXT
- __DATA
- __data
- __DATA
- __dyld
- __DATA

- __const
- __DATA
- __DATA
- __common
- __DATA
- __IMPORT
- __pointers
- __IMPORT
- _*jump*table
- __IMPORT
- __LINKEDIT
- /usr/lib/dyld
- /usr/lib/libncurses.5.4.dylib
- /usr/lib/libiconv.2.dylib
- /usr/lib/libSystem.B.dylib
- /usr/lib/libgcc_s.1.dylib
- t2
- Hu4F9u
- <\t'<]u1
- u
- <=tz<+u
- @ tc:/
- @$usr/
- <(t,<)tE
- <}te<]
- t(
- A`;A\t
- C0;C,u
- <.t0<\u
- tO
- debugger
- dump-po-strings
- dump-strings
- init-file
- noediting
- noprofile
- protected
- rcfile
- verbose
- version
- wordexp
- ~/.bashrc
- BASH_ENV
- run_wordexp
- --wordexp
- run*one*command
- I have no name!
- ??host??
- GNU bash, version %s-(%s)
- Usage:
- %s [GNU long option] [option] ...
- %s [GNU long option] [option] script-file ...
- GNU long options:

- Shell options:
- -irsD or -c command or -O shopt_option
- (invocation only)
- -%s or -o option
- Type `%s -c "help set"' for more information about shell options.
- Type `%s -c help' for more information about shell builtin commands.
- Use the `bashbug' command to report bugs.
- %s: option requires an argument
- %s: invalid option
- %c%c: invalid option
- Unix2003
- bin/sh
- POSIXLY_CORRECT
- POSIX_PEDANTIC
- \s-\v\$
- *BASH*IMPLICIT*DASH*PEE
- setuid(%u) failed: %s
- setreuid(%u,%u) failed: %s
- /etc/profile
- ~/.profile
- ~/.bash_profile
- ~/.bash_login
- /usr/local/share/bashdb/bashdb-main.inc
- FUNCNAME
- %s: is a directory
- %s: cannot execute binary file
- timed out waiting for input: auto-logout
- PROMPT_COMMAND
- reader_loop
- newline
- select
- function
- %a %b %d
- %H:%M:%S
- %I:%M:%S
- %I:%M %p
- syntax error near unexpected token `%s'
- syntax error near `%s'
- syntax error: unexpected end of file
- syntax error
- unexpected EOF while looking for matching `%c'
- readline stdin
- PIPESTATUS
- unexpected EOF while looking for `]]'
- syntax error in conditional expression: unexpected token `%s'
- syntax error in conditional expression
- unexpected token `%s', expected `)'
- expected `)'
- unexpected argument `%s' to conditional unary operator
- unexpected argument to conditional unary operator
- unexpected token `%s', conditional binary operator expected
- conditional binary operator expected
- unexpected argument `%s' to conditional binary operator

- unexpected argument to conditional binary operator
- unexpected token `%c' in conditional command
- unexpected token `%s' in conditional command
- unexpected token %d in conditional command
- unexpected EOF while looking for matching `)'
- logout
- Use "%s" to leave the shell.
- memory exhausted
- getcwd: cannot access parent directories
- /dev/tty
- `%s': not a valid identifier
- make*here*document: bad instruction type %d
- make_redirection: redirection instruction `%d' out of range
- clean*simple*command
- syntax error: arithmetic expression required
- syntax error: `;' unexpected
- syntax error: `((%s))'
- %s=(%s)
- for %s in
- select %s in
- case %s in
- cprintf: `%c': invalid format character
- <<%s%s
- <<< %s
- %d<&%d
- %d>&%d
- %d<&%s
- %d>&%s
- %d<&%d-
- %d>&%d-
- %d<&%s-
- %d>&%s-
- case %s in
- for ((
- print_command: bad connector `%d'
- function %s ()
- print_command
- dispose_command
- /dev/null
- cannot redirect standard input from /dev/null: %s
- execute*cond*node
- eval_builtin
- builtin_env
- return*temp*env
- %s: %s: bad interpreter
- cannot duplicate fd %d to fd %d
- real %2R
- user %2U
- sys %2S
- TIMEFORMAT
- TIMEFORMAT: `%c': invalid format character
- loop_redirections
- %s: readonly function

- execute_command
- function_calling
- execute-shell-function
- simple-command
- auto_resume
- substring
- saved redirects
- saved_redirects
- %s: restricted: cannot specify `/' in command names
- %s: command not found
- execute-command
- %*d%s%s
- pipe error
- pipe-file-descriptors
- execute_connection
- COMP_WORDBREAKS
- GLOBIGNORE
- HISTCONTROL
- HISTFILESIZE
- HISTIGNORE
- HISTSIZE
- HISTTIMEFORMAT
- IGNOREEOF
- LC_ALL
- LC_CTYPE
- LC_MESSAGES
- LC_NUMERIC
- MAILCHECK
- MAILPATH
- OPTERR
- OPTIND
- TERMCAP
- TERMINFO
- TEXTDOMAIN
- TEXTDOMAINDIR
- histchars
- ignoreeof
- all*local*variables: no function context at current scope
- pop*var*context: head of shell_variables not a function context
- pop*var*context: no global_variables context
- COLUMNS
- shell-init
- OLDPWD
- ignorespace
- ignoredups
- ignoreboth
- erasedups
- shell level (%d) too high, resetting to 1
- make*local*variable: no function context at current scope
- BASH_ARGV
- BASH_ARGC
- pop*scope: head of shell*variables not a temporary environment scope
- error importing function definition for `%s'

- /usr/gnu/bin:/usr/local/bin:/bin:/usr/bin:.
- HOSTTYPE
- darwin9.0
- OSTYPE
- MACHTYPE
- HOSTNAME
- BASH_VERSION
- BASH_VERSINFO
- BASH*EXECUTION*STRING
- ~/.sh_history
- ~/.bash_history
- SECONDS
- BASH_COMMAND
- BASH_SUBSHELL
- RANDOM
- LINENO
- HISTCMD
- DIRSTACK
- GROUPS
- BASH_SOURCE
- BASH_LINENO
- bash-maintainers@gnu.org
- unknown command error
- bad command type
- bad connector
- bad jump
- line
- %s:%s%d:
- (null)
- last command: %s
- Aborting...
- %s: warning:
- %s: %s:%s%d:
- %s: %s: %d
- %s: unbound variable
- %s%s%s: %s (error token is "%s")
- invalid number
- invalid arithmetic base
- value too great for base
- */%+-&^|
- syntax error: operand expected
- syntax error: invalid arithmetic operator
- identifier expected after pre-increment or pre-decrement
- missing `)'
- exponent less than 0
- division by 0
- expression expected
- `:' expected for conditional expression
- attempted assignment to non-variable
- bug: bad expassign token
- expression recursion level exceeded
- syntax error in expression
- recursion stack underflow

-
- Signal %d
- [%d]%c
- Stopped(%s)
- Stopped
- Running
- Done(%d)
- Exit %d
- Unknown status
- (core dumped)
- (wd: %s)
- [%ld: %d] tcsetattr
- [%d] %ld
- describe_pid: %ld: no such pid
- initialize*job*control: getpgrp failed
- initialize*job*control: setpgid
- no job control in this shell
- %s: line %d:
- (core dumped)
- (wd now: %s)
- notify*of*job_status
- SIGCHLD trap
- deleting stopped job %d with process group %ld
- job-working-directory
- wait_for: No record of process %ld
- %s: job has terminated
- %s: job %d already in background
- [%d]%s
- (wd: %s)
- wait*for*job: job %d is stopped
- wait: pid %ld is not a child of this shell
- child setpgid (%ld to %ld)
- forked pid %d appears in running job %d
- cannot make pipe for command substitution
- cannot make child for command substitution
- command_substitute: cannot duplicate pipe as fd 1
- command substitution
- %s: cannot assign list to array member
- bad substitution: no closing `%s' in %s
- cannot make pipe for process substitution
- cannot make child for process substitution
- cannot duplicate named pipe %s as fd %d
- process substitution
- bad substitution: no closing "`" in %s
- no match: %s
- %s: %s
- %s: parameter null or not set
- #%:-=?+/}
- %s: substring expression < 0
- %s: bad substitution
- $%s: cannot assign in this way
- You have mail in $_
- You have new mail in $_

- The mail in %s has been read
- SIGHUP
- SIGINT
- SIGQUIT
- SIGILL
- SIGTRAP
- SIGABRT
- SIGEMT
- SIGFPE
- SIGKILL
- SIGBUS
- SIGSEGV
- SIGSYS
- SIGPIPE
- SIGALRM
- SIGTERM
- SIGURG
- SIGSTOP
- SIGTSTP
- SIGCONT
- SIGCHLD
- SIGTTIN
- SIGTTOU
- SIGXCPU
- SIGXFSZ
- SIGVTALRM
- SIGPROF
- SIGWINCH
- SIGINFO
- SIGUSR1
- SIGUSR2
- RETURN
- invalid signal number
- exit trap
- interrupt trap
- debug trap
- error trap
- return trap
- run*pending*traps: bad value in trap_list[%d]: %p
- run*pending*traps: signal handler is SIG_DFL, resending %d (%s) to myself
- trap_handler: bad signal %d
- cannot allocate new file descriptor for bash input from fd %d
- save*bash*input: buffer already exists for new fd %d
- argument expected
- %s: integer expression expected
- %s: unary operator expected
- %s: binary operator expected
- `)' expected
- `)' expected, found %s
- missing `]'
- too many arguments
- @(#)Bash version 3.2.17(1) release GNU
- release

- %s.%d(%d)-%s
- %s.%d(%d)
- i386-apple-darwin9.0
- GNU bash, version %s (%s)
- Copyright (C) 2005 Free Software Foundation, Inc.
- bad array subscript
- array assign
- %s[%s: %s
- [%s]: %s
- %s: cannot assign to non-numeric index
- ;&()|<>
- %s: cannot create: %s
- %s%s%s
- FIGNORE
- "'><=;|&(:
- "'@><=;|&(:
- comment-begin
- shell-expand-line
- history-expand-line
- magic-space
- alias-expand-line
- history-and-alias-expand-line
- insert-last-argument
- operate-and-get-next
- display-shell-version
- edit-and-execute-command
- complete-into-braces
- complete-filename
- possible-filename-completions
- complete-username
- possible-username-completions
- complete-hostname
- possible-hostname-completions
- complete-variable
- possible-variable-completions
- complete-command
- possible-command-completions
- glob-complete-word
- glob-expand-word
- glob-list-expansions
- dynamic-complete-history
- \"'@<>=;|&()#$`?*[!:{
- $include
- HOSTFILE
- hostname*completion*file
- /etc/hosts
- C-xC-e
- fc -e vi
- fc -e "${VISUAL:-${EDITOR:-vi}}"
- fc -e "${VISUAL:-${EDITOR:-emacs}}"
- ;|&{(`
- symlink-hook
- bash*execute*unix_command: cannot find keymap for command

- bash*execute*unix_command
- %s: first non-whitespace character is not `"'
- no closing `%c' in %s
- %s: missing colon separator
- /usr/share/locale
- ()<>;&|
- LC_COLLATE
- LC_TIME
- #: %s:%d
- msgid %s%s
- msgstr ""
- /dev/tcp//
- /dev/udp//
- file descriptor out of range
- %s: ambiguous redirect
- %s: cannot overwrite existing file
- %s: restricted: cannot redirect output
- cannot create temp file for here document: %s
- redirection error: cannot duplicate fd
- sh-thd
- COMP_LINE
- COMP_POINT
- COMP_WORDS
- COMP_CWORD
- completion: function `%s' not found
- COMPREPLY
- progcomp_insert: %s: NULL COMPSPEC
- xmalloc: cannot allocate %lu bytes (%lu bytes allocated)
- xrealloc: cannot reallocate %lu bytes (%lu bytes allocated)
- Display the possible completions depending on the options. Intended
- to be used from within a shell function generating possible completions.
- If the optional WORD argument is supplied, matches against WORD are
- generated.
- For each NAME, specify how arguments are to be completed.
- If the -p option is supplied, or if no options are supplied, existing
- completion specifications are printed in a way that allows them to be
- reused as input. The -r option removes a completion specification for
- each NAME, or, if no NAMEs are supplied, all completion specifications.
- printf formats and prints ARGUMENTS under control of the FORMAT. FORMAT
- is a character string which contains three types of objects: plain
- characters, which are simply copied to standard output, character escape
- sequences which are converted and copied to the standard output, and
- format specifications, each of which causes printing of the next successive
- argument. In addition to the standard printf(1) formats, %b means to
- expand backslash escape sequences in the corresponding argument, and %q
- means to quote the argument in a way that can be reused as shell input.
- If the -v option is supplied, the output is placed into the value of the
- shell variable VAR rather than being sent to the standard output.
- Toggle the values of variables controlling optional behavior.
- The -s flag means to enable (set) each OPTNAME; the -u flag
- unsets each OPTNAME. The -q flag suppresses output; the exit
- status indicates whether each OPTNAME is set or unset. The -o
- option restricts the OPTNAMEs to those defined for use with

- `set -o'. With no options, or with the -p option, a list of all
- settable options is displayed, with an indication of whether or
- not each is set.
- Display the list of currently remembered directories. Directories
- find their way onto the list with the `pushd' command; you can get
- back up through the list with the `popd' command.
- The -l flag specifies that `dirs' should not print shorthand versions
- of directories which are relative to your home directory. This means
- that `~/bin' might be displayed as`/homes/bfox/bin'. The -v flag
- causes `dirs' to print the directory stack with one entry per line,
- prepending the directory name with its position in the stack. The -p
- flag does the same thing, but the stack position is not prepended.
- The -c flag clears the directory stack by deleting all of the elements.
- +N
- displays the Nth entry counting from the left of the list shown by
- dirs when invoked without options, starting with zero.
- -N
- displays the Nth entry counting from the right of the list shown by
- dirs when invoked without options, starting with zero.
- Removes entries from the directory stack. With no arguments,
- removes the top directory from the stack, and cd's to the new
- top directory.
- +N
- removes the Nth entry counting from the left of the list
- shown by `dirs', starting with zero. For example:`popd +0'
- removes the first directory, `popd +1' the second.
- -N
- removes the Nth entry counting from the right of the list
- shown by `dirs', starting with zero. For example:`popd -0'
- removes the last directory, `popd -1' the next to last.
- -n
- suppress the normal change of directory when removing directories
- from the stack, so only the stack is manipulated.
- You can see the directory stack with the `dirs' command.
- Adds a directory to the top of the directory stack, or rotates
- the stack, making the new top of the stack the current working
- directory. With no arguments, exchanges the top two directories.
- +N
- Rotates the stack so that the Nth directory (counting
- from the left of the list shown by `dirs', starting with
- zero) is at the top.
- -N
- Rotates the stack so that the Nth directory (counting
- from the right of the list shown by `dirs', starting with
- zero) is at the top.
- -n
- suppress the normal change of directory when adding directories
- to the stack, so only the stack is manipulated.
- dir
- adds DIR to the directory stack at the top, making it the
- new current working directory.
- You can see the directory stack with the `dirs' command.
- BASH_VERSION

- Version information for this Bash.
- CDPATH
- A colon-separated list of directories to search
- for directries given as arguments to `cd'.
- GLOBIGNORE
- A colon-separated list of patterns describing filenames to
- be ignored by pathname expansion.
- HISTFILE
- The name of the file where your command history is stored.
- HISTFILESIZE
- The maximum number of lines this file can contain.
- HISTSIZE
- The maximum number of history lines that a running
- shell can access.
- HOME
- The complete pathname to your login directory.
- HOSTNAME
- The name of the current host.
- HOSTTYPE
- The type of CPU this version of Bash is running under.
- IGNOREEOF
- Controls the action of the shell on receipt of an EOF
- character as the sole input. If set, then the value
- of it is the number of EOF characters that can be seen
- in a row on an empty line before the shell will exit
- (default 10). When unset, EOF signifies the end of input.
- MACHTYPE
- A string describing the current system Bash is running on.
- MAILCHECK
- How often, in seconds, Bash checks for new mail.
- MAILPATH
- A colon-separated list of filenames which Bash checks
- for new mail.
- OSTYPE
- The version of Unix this version of Bash is running on.
- PATH
- A colon-separated list of directories to search when
- looking for commands.
- PROMPT_COMMAND
- A command to be executed before the printing of each
- primary prompt.
- PS1
- The primary prompt string.
- PS2
- The secondary prompt string.
- PWD
- The full pathname of the current directory.
- SHELLOPTS
- A colon-separated list of enabled shell options.
- TERM
- The name of the current terminal type.
- TIMEFORMAT
- The output format for timing statistics displayed by the

- `time' reserved word.
- auto_resume
- Non-null means a command word appearing on a line by
- itself is first looked for in the list of currently
- stopped jobs. If found there, that job is foregrounded.
- A value of `exact' means that the command word must
- exactly match a command in the list of stopped jobs. A
- value of `substring' means that the command word must
- match a substring of the job. Any other value means that
- the command must be a prefix of a stopped job.
- histchars
- Characters controlling history expansion and quick
- substitution. The first character is the history
- substitution character, usually `!'. The second is
- the `quick substitution' character, usually `^'. The
- third is the `history comment' character, usually `#'.
- HISTIGNORE
- A colon-separated list of patterns used to decide which
- commands should be saved on the history list.
- Returns a status of 0 or 1 depending on the evaluation of the conditional
- expression EXPRESSION. Expressions are composed of the same primaries used
- by the `test' builtin, and may be combined using the following operators
- (EXPRESSION)
- Returns the value of EXPRESSION
- ! EXPRESSION
- True if EXPRESSION is false; else false
- EXPR1 && EXPR2
- True if both EXPR1 and EXPR2 are true; else false
- EXPR1 || EXPR2
- True if either EXPR1 or EXPR2 is true; else false
- When the `==' and `!=' operators are used, the string to the right of the
- operator is used as a pattern and pattern matching is performed. The
- && and || operators do not evaluate EXPR2 if EXPR1 is sufficient to
- determine the expression's value.
- The EXPRESSION is evaluated according to the rules for arithmetic
- evaluation. Equivalent to "let EXPRESSION".
- Equivalent to the JOB_SPEC argument to the `fg' command. Resume a
- stopped or background job. JOB_SPEC can specify either a job name
- or a job number. Following JOB_SPEC with a `&' places the job in
- the background, as if the job specification had been supplied as an
- argument to `bg'.
- Run a set of commands in a group. This is one way to redirect an
- entire set of commands.
- Create a simple command invoked by NAME which runs COMMANDS.
- Arguments on the command line along with NAME are passed to the
- function as $0 .. $n.
- Expand and execute COMMANDS as long as the final command in the
- `until' COMMANDS has an exit status which is not zero.
- Expand and execute COMMANDS as long as the final command in the
- `while' COMMANDS has an exit status of zero.
- The `if COMMANDS' list is executed. If its exit status is zero, then the
- `then COMMANDS' list is executed. Otherwise, each `elif COMMANDS' list is
- executed in turn, and if its exit status is zero, the corresponding

- `then COMMANDS' list is executed and the if command completes. Otherwise,
- the `else COMMANDS' list is executed, if present. The exit status of the
- entire construct is the exit status of the last command executed, or zero
- if no condition tested true.
- Selectively execute COMMANDS based upon WORD matching PATTERN. The
- `|' is used to separate multiple patterns.
- Execute PIPELINE and print a summary of the real time, user CPU time,
- and system CPU time spent executing PIPELINE when it terminates.
- The return status is the return status of PIPELINE. The `-p' option
- prints the timing summary in a slightly different format. This uses
- the value of the TIMEFORMAT variable as the output format.
- The WORDS are expanded, generating a list of words. The
- set of expanded words is printed on the standard error, each
- preceded by a number. If `in WORDS' is not present,` in "$@"
- is assumed. The PS3 prompt is then displayed and a line read
- from the standard input. If the line consists of the number
- corresponding to one of the displayed words, then NAME is set
- to that word. If the line is empty, WORDS and the prompt are
- redisplayed. If EOF is read, the command completes. Any other
- value read causes NAME to be set to null. The line read is saved
- in the variable REPLY. COMMANDS are executed after each selection
- until a break command is executed.
- Equivalent to
- ((EXP1))
- while ((EXP2)); do
- COMMANDS
- ((EXP3))
- EXP1, EXP2, and EXP3 are arithmetic expressions. If any expression is
- omitted, it behaves as if it evaluates to 1.
- The `for' loop executes a sequence of commands for each member in a
- list of items. If `in WORDS ...;' is not present, then` in "$@" is
- assumed. For each element in WORDS, NAME is set to that element, and
- the COMMANDS are executed.
- Wait for the specified process and report its termination status. If
- N is not given, all currently active child processes are waited for,
- and the return code is zero. N may be a process ID or a job
- specification; if a job spec is given, all processes in the job's
- pipeline are waited for.
- The user file-creation mask is set to MODE. If MODE is omitted, or if
- `-S' is supplied, the current value of the mask is printed. The` -S'
- option makes the output symbolic; otherwise an octal number is output.
- If `-p' is supplied, and MODE is omitted, the output is in a form
- that may be used as input. If MODE begins with a digit, it is
- interpreted as an octal number, otherwise it is a symbolic mode string
- like that accepted by chmod(1).
- Ulimit provides control over the resources available to processes
- started by the shell, on systems that allow such control. If an
- option is given, it is interpreted as follows:
- -S
- use the `soft' resource limit
- -H
- use the `hard' resource limit
- -a

- all current limits are reported
- -c
- the maximum size of core files created
- -d
- the maximum size of a process's data segment
- -e
- the maximum scheduling priority (`nice')
- -f
- the maximum size of files written by the shell and its children
- -i
- the maximum number of pending signals
- -l
- the maximum size a process may lock into memory
- -m
- the maximum resident set size
- -n
- the maximum number of open file descriptors
- -p
- the pipe buffer size
- -q
- the maximum number of bytes in POSIX message queues
- -r
- the maximum real-time scheduling priority
- -s
- the maximum stack size
- -t
- the maximum amount of cpu time in seconds
- -u
- the maximum number of user processes
- -v
- the size of virtual memory
- -x
- the maximum number of file locks
- If LIMIT is given, it is the new value of the specified resource;
- the special LIMIT values `soft',` hard', and `unlimited' stand for
- the current soft limit, the current hard limit, and no limit, respectively.
- Otherwise, the current value of the specified resource is printed.
- If no option is given, then -f is assumed. Values are in 1024-byte
- increments, except for -t, which is in seconds, -p, which is in
- increments of 512 bytes, and -u, which is an unscaled number of
- processes.
- For each NAME, indicate how it would be interpreted if used as a
- command name.
- If the -t option is used, `type' outputs a single word which is one of
- `alias',` keyword', `function',` builtin', `file' or`', if NAME is an
- alias, shell reserved word, shell function, shell builtin, disk file,
- or unfound, respectively.
- If the -p flag is used, `type' either returns the name of the disk
- file that would be executed, or nothing if `type -t NAME' would not
- return `file'.
- If the -a flag is used, `type' displays all of the places that contain
- an executable named `file'. This includes aliases, builtins, and
- functions, if and only if the -p flag is not also used.

- The -f flag suppresses shell function lookup.
- The -P flag forces a PATH search for each NAME, even if it is an alias,
- builtin, or function, and returns the name of the disk file that would
- be executed.
- The command ARG is to be read and executed when the shell receives
- signal(s) SIGNAL*SPEC. If ARG is absent (and a single SIGNAL*SPEC
- is supplied) or `-', each specified signal is reset to its original
- value. If ARG is the null string each SIGNAL_SPEC is ignored by the
- shell and by the commands it invokes. If a SIGNAL_SPEC is EXIT (0)
- the command ARG is executed on exit from the shell. If a SIGNAL_SPEC
- is DEBUG, ARG is executed after every simple command. If the`-p' option
- is supplied then the trap commands associated with each SIGNAL_SPEC are
- displayed. If no arguments are supplied or if only `-p' is given, trap
- prints the list of commands associated with each signal. Each SIGNAL_SPEC
- is either a signal name in or a signal number. Signal names
- are case insensitive and the SIG prefix is optional. `trap -l' prints
- a list of signal names and their corresponding numbers. Note that a
- signal can be sent to the shell with "kill -signal $$".
- Print the accumulated user and system times for processes run from
- the shell.
- This is a synonym for the "test" builtin, but the last
- argument must be a literal `]', to match the opening`['.
- Exits with a status of 0 (true) or 1 (false) depending on
- the evaluation of EXPR. Expressions may be unary or binary. Unary
- expressions are often used to examine the status of a file. There
- are string operators as well, and numeric comparison operators.
- File operators:
- -a FILE True if file exists.
- -b FILE True if file is block special.
- -c FILE True if file is character special.
- -d FILE True if file is a directory.
- -e FILE True if file exists.
- -f FILE True if file exists and is a regular file.
- -g FILE True if file is set-group-id.
- -h FILE True if file is a symbolic link.
- -L FILE True if file is a symbolic link.
- -k FILE True if file has its `sticky' bit set.
- -p FILE True if file is a named pipe.
- -r FILE True if file is readable by you.
- -s FILE True if file exists and is not empty.
- -S FILE True if file is a socket.
- -t FD True if FD is opened on a terminal.
- -u FILE True if the file is set-user-id.
- -w FILE True if the file is writable by you.
- -x FILE True if the file is executable by you.
- -O FILE True if the file is effectively owned by you.
- -G FILE True if the file is effectively owned by your group.
- -N FILE True if the file has been modified since it was last read.
- FILE1 -nt FILE2 True if file1 is newer than file2 (according to
- modification date).
- FILE1 -ot FILE2 True if file1 is older than file2.
- FILE1 -ef FILE2 True if file1 is a hard link to file2.
- String operators:

- -z STRING True if string is empty.
- -n STRING
- STRING True if string is not empty.
- STRING1 = STRING2
- True if the strings are equal.
- STRING1 != STRING2
- True if the strings are not equal.
- STRING1 < STRING2
- True if STRING1 sorts before STRING2 lexicographically.
- STRING1 > STRING2
- True if STRING1 sorts after STRING2 lexicographically.
- Other operators:
- -o OPTION True if the shell option OPTION is enabled.
- ! EXPR True if expr is false.
- EXPR1 -a EXPR2 True if both expr1 AND expr2 are true.
- EXPR1 -o EXPR2 True if either expr1 OR expr2 is true.
- arg1 OP arg2 Arithmetic tests. OP is one of -eq, -ne,
- -lt, -le, -gt, or -ge.
- Arithmetic binary operators return true if ARG1 is equal, not-equal,
- less-than, less-than-or-equal, greater-than, or greater-than-or-equal
- than ARG2.
- Suspend the execution of this shell until it receives a SIGCONT
- signal. The `-f' if specified says not to complain about this
- being a login shell if it is; just suspend anyway.
- Read and execute commands from FILENAME and return. The pathnames
- in $PATH are used to find the directory containing FILENAME. If any
- ARGUMENTS are supplied, they become the positional parameters when
- FILENAME is executed.
- The positional parameters from $N+1 ... are renamed to $1 ... If N is
- not given, it is assumed to be 1.
- The given NAMEs are marked readonly and the values of these NAMEs may
- not be changed by subsequent assignment. If the -f option is given,
- then functions corresponding to the NAMEs are so marked. If no
- arguments are given, or if `-p' is given, a list of all readonly names
- is printed. The `-a' option means to treat each NAME as
- an array variable. An argument of `--' disables further option
- processing.
- NAMEs are marked for automatic export to the environment of
- subsequently executed commands. If the -f option is given,
- the NAMEs refer to functions. If no NAMEs are given, or if `-p'
- is given, a list of all names that are exported in this shell is
- printed. An argument of `-n' says to remove the export property
- from subsequent NAMEs. An argument of `--' disables further option
- processing.
- For each NAME, remove the corresponding variable or function. Given
- the `-v', unset will only act on variables. Given the `-f' flag,
- unset will only act on functions. With neither flag, unset first
- tries to unset a variable, and if that fails, then tries to unset a
- function. Some variables cannot be unset; also see readonly.
- -a Mark variables which are modified or created for export.
- -b Notify of job termination immediately.
- -e Exit immediately if a command exits with a non-zero status.
- -f Disable file name generation (globbing).

- -h Remember the location of commands as they are looked up.
- -k All assignment arguments are placed in the environment for a
- command, not just those that precede the command name.
- -m Job control is enabled.
- -n Read commands but do not execute them.
- -o option-name
- Set the variable corresponding to option-name:
- allexport same as -a
- braceexpand same as -B
- emacs use an emacs-style line editing interface
- errexit same as -e
- errtrace same as -E
- functrace same as -T
- hashall same as -h
- histexpand same as -H
- history enable command history
- ignoreeof the shell will not exit upon reading EOF
- interactive-comments
- allow comments to appear in interactive commands
- keyword same as -k
- monitor same as -m
- noclobber same as -C
- noexec same as -n
- noglob same as -f
- nolog currently accepted but ignored
- notify same as -b
- nounset same as -u
- onecmd same as -t
- physical same as -P
- pipefail the return value of a pipeline is the status of
- the last command to exit with a non-zero status,
- or zero if no command exited with a non-zero status
- posix change the behavior of bash where the default
- operation differs from the 1003.2 standard to
- match the standard
- privileged same as -p
- verbose same as -v
- vi use a vi-style line editing interface
- xtrace same as -x
- -p Turned on whenever the real and effective user ids do not match.
- Disables processing of the $ENV file and importing of shell
- functions. Turning this option off causes the effective uid and
- gid to be set to the real uid and gid.
- -t Exit after reading and executing one command.
- -u Treat unset variables as an error when substituting.
- -v Print shell input lines as they are read.
- -x Print commands and their arguments as they are executed.
- -B the shell will perform brace expansion
- -C If set, disallow existing regular files to be overwritten
- by redirection of output.
- -E If set, the ERR trap is inherited by shell functions.
- -H Enable ! style history substitution. This flag is on
- by default when the shell is interactive.

- -P If set, do not follow symbolic links when executing commands
- such as cd which change the current directory.
- -T If set, the DEBUG trap is inherited by shell functions.
-
    - Assign any remaining arguments to the positional parameters.

- The -x and -v options are turned off.
- Using + rather than - causes these flags to be turned off. The
- flags can also be used upon invocation of the shell. The current
- set of flags may be found in $-. The remaining n ARGs are positional
- parameters and are assigned, in order, to $1, $2, .. $n. If no
- ARGs are given, all shell variables are printed.
- Causes a function to exit with the return value specified by N. If N
- is omitted, the return status is that of the last command.
- One line is read from the standard input, or from file descriptor FD if the
- -u option is supplied, and the first word is assigned to the first NAME,
- the second word to the second NAME, and so on, with leftover words assigned
- to the last NAME. Only the characters found in $IFS are recognized as word
- delimiters. If no NAMEs are supplied, the line read is stored in the REPLY
- variable. If the -r option is given, this signifies raw' input, and backslash escaping is disabled. The -d option causes read to continue until the first character of DELIM is read, rather than newline. If the -p option is supplied, the string PROMPT is output without a trailing newline before attempting to read. If -a is supplied, the words read are assigned to sequential indices of ARRAY, starting at zero. If -e is supplied and the shell is interactive, readline is used to obtain the line. If -n is supplied with a non-zero NCHARS argument, read returns after NCHARS characters have been read. The -s option causes input coming from a terminal to not be echoed. The -t option causes read to time out and return failure if a complete line of input is not read within TIMEOUT seconds. If the TMOUT variable is set, its value is the default timeout. The return code is zero, unless end-of-file is encountered, read times out, or an invalid file descriptor is supplied as the argument to -u. Each ARG is an arithmetic expression to be evaluated. Evaluation is done in fixed-width integers with no check for overflow, though division by 0 is trapped and flagged as an error. The following list of operators is grouped into levels of equal-precedence operators. The levels are listed in order of decreasing precedence. id++, id-- variable post-increment, post-decrement ++id, --id variable pre-increment, pre-decrement unary minus, plus logical and bitwise negation exponentiation *, /, % multiplication, division, remainder addition, subtraction <<, >> left and right bitwise shifts <=, >=, <, > comparison ==, != equality, inequality bitwise AND bitwise XOR bitwise OR logical AND logical OR expr ? expr : expr conditional operator =, *=, /=, %=, +=, -=, <<=, >>=, &=, ^=, |= assignment Shell variables are allowed as operands. The name of the variable is replaced by its value (coerced to a fixed-width integer) within an expression. The variable need not have its integer attribute turned on to be used in an expression. Operators are evaluated in order of precedence. Sub-expressions in parentheses are evaluated first and may override the precedence rules above. If the last ARG evaluates to 0, let returns 1; 0 is returned otherwise. Send the processes named by PID (or JOBSPEC) the signal SIGSPEC. If SIGSPEC is not present, then SIGTERM is assumed. An argument of -l'
- lists the signal names; if arguments follow -l' they are assumed to be signal numbers for which names should be listed. Kill is a shell builtin for two reasons: it allows job IDs to be used instead of process IDs, and, if you have reached the limit on processes that you can create, you don't have to start a process to kill another one. By default, removes each JOBSPEC argument from the table of active jobs. If the -h option is given, the job is not removed from the table, but is marked so that SIGHUP is not sent to the job if the shell receives a SIGHUP. The -a option, when JOBSPEC is not supplied, means to remove all jobs from the job table; the -r option

means to remove only running jobs. Lists the active jobs. The -l option lists process id's in addition to the normal information; the -p option lists process id's only. If -n is given, only processes that have changed status since the last notification are printed. JOBSPEC restricts output to that job. The -r and -s options restrict output to running and stopped jobs only, respectively. Without options, the status of all active jobs is printed. If -x is given, COMMAND is run after all job specifications that appear in ARGS have been replaced with the process ID of that job's process group leader. Display the history list with line numbers. Lines listed with with a *' have been modified. Argument of N says to list only*

- *the last N lines. The* *-c' option causes the history list to be cleared by deleting all of the entries. The* *-d' option deletes*
- *the history entry at offset OFFSET. The* *-w' option writes out the current history to the history file;* *-r' means to read the file and*
- *append the contents to the history list instead.* *-a' means to append history lines from this session to the history file. Argument* *-n' means to read all history lines not already read*
- *from the history file and append them to the history list.*
- *If FILENAME is given, then that is used as the history file else*
- *if $HISTFILE has a value, that is used, else ~/.bash_history.*
- *If the -s option is supplied, the non-option ARGs are appended to*
- *the history list as a single entry. The -p option means to perform*
- *history expansion on each ARG and display the result, without storing*
- *anything in the history list.*
- *If the $HISTTIMEFORMAT variable is set and not null, its value is used*
- *as a format string for strftime(3) to print the time stamp associated*
- *with each displayed history entry. No time stamps are printed otherwise.*
- *Display helpful information about builtin commands. If PATTERN is*
- *specified, gives detailed help on all commands matching PATTERN,*
- *otherwise a list of the builtins is printed. The -s option*
- *restricts the output for each builtin command matching PATTERN to*
- *a short usage synopsis.*
- *For each NAME, the full pathname of the command is determined and*
- *remembered. If the -p option is supplied, PATHNAME is used as the*
- *full pathname of NAME, and no path search is performed. The -r*
- *option causes the shell to forget all remembered locations. The -d*
- *option causes the shell to forget the remembered location of each NAME.*
- *If the -t option is supplied the full pathname to which each NAME*
- *corresponds is printed. If multiple NAME arguments are supplied with*
- *-t, the NAME is printed before the hashed full pathname. The -l option*
- *causes output to be displayed in a format that may be reused as input.*
- *If no arguments are given, information about remembered commands is displayed.*
- *Place each JOB_SPEC in the background, as if it had been started with*
- *&'. If JOB_SPEC is not present, the shell's notion of the current job is used. Place JOB_SPEC in the foreground, and make it the current job. If JOB_SPEC is not present, the shell's notion of the current job is used. fc is used to list or edit and re-execute commands from the history list. FIRST and LAST can be numbers specifying the range, or FIRST can be a string, which means the most recent command beginning with that string. -e ENAME selects which editor to use. Default is FCEDIT, then EDITOR, then vi. -l means list lines instead of editing. -n means no line numbers listed. -r means reverse the order of the lines (making it newest listed first). With the* *fc -s [pat=rep ...] [command]' format, the command is*
- *re-executed after the substitution OLD=NEW is performed.*
- *A useful alias to use with this is r='fc -s', so that typing* *r cc' runs the last command beginning with <code>cc' and typing</code>r' re-executes the last command. Logout of a login shell. Exit the shell with a status of N. If N is omitted, the exit status is that of the last command executed. Exec FILE, replacing this shell with the specified program. If FILE is not*

specified, the redirections take effect in this shell. If the first argument is -l', then place a dash in the

- zeroth arg passed to FILE, as login does. If the -c' option is supplied, FILE is executed with a null environment. The -a'
- option means to make set argv[0] of the executed process to NAME.
- If the file cannot be executed and the shell is not interactive,
- then the shell exits, unless the shell option execfail' is set. Getopts is used by shell procedures to parse positional parameters. OPTSTRING contains the option letters to be recognized; if a letter is followed by a colon, the option is expected to have an argument, which should be separated from it by white space. Each time it is invoked, getopts will place the next option in the shell variable $name, initializing name if it does not exist, and the index of the next argument to be processed into the shell variable OPTIND. OPTIND is initialized to 1 each time the shell or a shell script is invoked. When an option requires an argument, getopts places that argument into the shell variable OPTARG. getopts reports errors in one of two ways. If the first character of OPTSTRING is a colon, getopts uses silent error reporting. In this mode, no error messages are printed. If an invalid option is seen, getopts places the option character found into OPTARG. If a required argument is not found, getopts places a ':' into NAME and sets OPTARG to the option character found. If getopts is not in silent mode, and an invalid option is seen, getopts places '?' into NAME and unsets OPTARG. If a required argument is not found, a '?' is placed in NAME, OPTARG is unset, and a diagnostic message is printed. If the shell variable OPTERR has the value 0, getopts disables the printing of error messages, even if the first character of OPTSTRING is not a colon. OPTERR has the value 1 by default. Getopts normally parses the positional parameters ($0 - $9), but if more arguments are given, they are parsed instead. Read ARGs as input to the shell and execute the resulting command(s). Enable and disable builtin shell commands. This allows you to use a disk command which has the same name as a shell builtin without specifying a full pathname. If -n is used, the NAMEs become disabled; otherwise NAMEs are enabled. For example, to use the test' found in $PATH instead of the shell builtin
- version, type enable -n test'. On systems supporting dynamic loading, the -f option may be used to load new builtins from the shared object FILENAME. The -d option will delete a builtin previously loaded with -f. If no non-option names are given, or the -p option is supplied, a list of builtins is printed. The -a option means to print every builtin with an indication of whether or not it is enabled. The -s option restricts the output to the POSIX.2 special' builtins. The -n option displays a list of all disabled builtins.
- Output the ARGs. If -n is specified, the trailing newline is
- suppressed. If the -e option is given, interpretation of the
- following backslash-escaped characters is turned on:
- alert (bell)
- backspace
- suppress trailing newline
- escape character
- form feed
- newline
- carriage return
- horizontal tab
- vertical tab
- backslash
- the character whose ASCII code is NNN (octal). NNN can be
- 0 to 3 octal digits
- You can explicitly turn off the interpretation of the above characters
- with the -E option.
- Create a local variable called NAME, and give it VALUE. LOCAL
- can only be used within a function; it makes the variable NAME

- *have a visible scope restricted to that function and its children.*
- *Obsolete. See* declare'. Declare variables and/or give them attributes. If no NAMEs are given, then display the values of variables instead. The -p option will display the attributes and values of each NAME. The flags are: -a to make NAMEs arrays (if supported) -f to select from among function names only -F to display function names (and line number and source file name if debugging) without definitions -i to make NAMEs have the *integer' attribute*
- *-r*
- *to make NAMEs readonly*
- *-t*
- *to make NAMEs have the* trace' attribute -x to make NAMEs export Variables with the integer attribute have arithmetic evaluation (see *let') done when the variable is assigned to.*
- *When displaying values of variables, -f displays a function's name*
- *and definition. The -F option restricts the display to function*
- *name only.*
- *Using* +' instead of*-' turns off the given attribute instead. When*
- *used in a function, makes NAMEs local, as with the* local' command. Runs COMMAND with ARGS ignoring shell functions. If you have a shell function called <code>ls', and you wish to call the command</code>ls', you can say "command ls". If the -p option is given, a default value is used for PATH that is guaranteed to find all of the standard utilities. If the -V or -v option is given, a string is printed describing COMMAND. The -V option produces a more verbose description. Return an unsuccessful result. Return a successful result. No effect; the command does nothing. A zero exit code is returned. Print the current working directory. With the -P option, pwd prints the physical directory, without any symbolic links; the -L option makes pwd follow symbolic links. Change the current directory to DIR. The variable $HOME is the default DIR. The variable CDPATH defines the search path for the directory containing DIR. Alternative directory names in CDPATH are separated by a colon (:). A null directory name is the same as the current directory, i.e. *.'. If DIR begins with a slash (/),*
- *then CDPATH is not used. If the directory is not found, and the*
- *shell option* cdable_vars' is set, then try the word as a variable name. If that variable has a value, then cd to the value of that variable. The -P option says to use the physical directory structure instead of following symbolic links; the -L option forces symbolic links to be followed. Returns the context of the current subroutine call. Without EXPR, returns "$line $filename". With EXPR, returns "$line $subroutine $filename"; this extra information can be used to provide a stack trace. The value of EXPR indicates how many call frames to go back before the current one; the top frame is frame 0. Run a shell builtin. This is useful when you wish to rename a shell builtin to be a function, but need the functionality of the builtin within the function itself. Resume the next iteration of the enclosing FOR, WHILE or UNTIL loop. If N is specified, resume at the N-th enclosing loop. Exit from within a FOR, WHILE or UNTIL loop. If N is specified, break N levels. Bind a key sequence to a Readline function or a macro, or set a Readline variable. The non-option argument syntax is equivalent to that found in ~/.inputrc, but must be passed as a single argument: bind '"\C-x\C-r": re-read-init-file'. bind accepts the following options: -m keymap Use *keymap' as the keymap for the duration of this*
- *command. Acceptable keymap names are emacs,*
- *emacs-standard, emacs-meta, emacs-ctlx, vi, vi-move,*
- *vi-command, and vi-insert.*
- *-l List names of functions.*
- *-P List function names and bindings.*
- *-p List functions and bindings in a form that can be*
- *reused as input.*
- *-r keyseq Remove the binding for KEYSEQ.*
- *-x keyseq:shell-command*

- *Cause SHELL-COMMAND to be executed when*
- *KEYSEQ is entered.*
- *-f filename Read key bindings from FILENAME.*
- *-q function-name Query about which keys invoke the named function.*
- *-u function-name Unbind all keys which are bound to the named function.*
- *-V List variable names and values*
- *-v List variable names and values in a form that can*
- *be reused as input.*
- *-S List key sequences that invoke macros and their values*
- *-s List key sequences that invoke macros and their values*
- *in a form that can be reused as input.*
- *Remove NAMEs from the list of defined aliases. If the -a option is given,*
- *then remove all alias definitions.*
- alias' with no arguments or with the -p option prints the list of aliases in the form alias NAME=VALUE on standard output. Otherwise, an alias is defined for each NAME whose VALUE is given. A trailing space in VALUE causes the next word to be checked for alias substitution when the alias is expanded. Alias returns true unless a NAME is given for which no alias has been defined. alias [-p] [name[=value] ...] unalias unalias [-a] name [name ...] bind [-lpvsPVS] [-m keymap] [-f filename] [-q name] [-u name] [-r keyseq] [-x keyseq:shell-command] break [n] continue continue [n] builtin builtin [shell-builtin [arg ...]] caller caller [EXPR] cd [-L|-P] [dir] pwd [-LP] command command [-pVv] command [arg ...] declare declare [-afFirtx] [-p] [name[=value] ...] typeset typeset [-afFirtx] [-p] name[=value] ... local name[=value] ... echo [-neE] [arg ...] enable enable [-pnds] [-a] [-f filename] [name ...] eval [arg ...] getopts getopts optstring name [arg] exec [-cl] [-a name] file [redirection ...] exit [n] fc [-e ename] [-nlr] [first] [last] or fc -s [pat=rep] [cmd] fg [job_spec] bg [job_spec ...] hash [-lr] [-p pathname] [-dt] [name ...] help [-s] [pattern ...] history [-c] [-d offset] [n] or history -awrn [filename] or history -ps arg [arg...] jobs [-lnprs] [jobspec ...] or jobs -x command [args] disown disown [-h] [-ar] [jobspec ...] kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec] let arg [arg ...] read [-ers] [-u fd] [-t timeout] [-p prompt] [-a array] [-n nchars] [-d delim] [name ...] return return [n] set [--abefhkmnptuvxBCHP] [-o option] [arg ...] unset [-f] [-v] [name ...] export [-nf] [name[=value] ...] or export -p readonly readonly [-af] [name[=value] ...] or readonly -p shift [n] source filename [arguments] . filename [arguments] suspend suspend [-f] test [expr] [arg...] trap [-lp] [arg signal_spec ...] type [-afptP] name [name ...] ulimit ulimit [-SHacdfilmnpqstuvx] [limit] umask [-p] [-S] [mode] wait [n] for NAME [in WORDS ... ;] do COMMANDS; done for ((exp1; exp2; exp3)); do COMMANDS; done select NAME [in WORDS ... ;] do COMMANDS; done time [-p] PIPELINE case WORD in [PATTERN [| PATTERN]...) COMMANDS ;;]... esac if COMMANDS; then COMMANDS; [elif COMMANDS; then COMMANDS;]... [else COMMANDS;] fi while COMMANDS; do COMMANDS; done until COMMANDS; do COMMANDS; done function NAME { COMMANDS ; } or NAME () { COMMANDS ; } { ... } { COMMANDS ; } JOB_SPEC [&] ((...)) ((expression)) [[...]] [[expression]] variables variables - Some variable names and meanings pushd [dir | +N | -N] [-n] popd [+N | -N] [-n] dirs [-clpv] [+N] [-N] shopt [-pqsu] [-o long-option] optname [optname...] printf printf [-v var] format [arguments] complete complete [-abcdefgjksuv] [-pr] [-o option] [-A action] [-G globpat] [-W wordlist] [-P prefix] [-S su compgen [-abcdefgjksuv] [-o option] [-A action] [-G globpat] [-W wordlist] [-P prefix] [-S suffix] [alias *%s': invalid alias name*
- *bind_builtin*
- *lvpVPsSf:q:u:m:r:x:*
- %s': invalid keymap name %s: cannot read: %s *%s': unknown function name*

- %s is not bound to any keys.
- %s can be invoked via
- %s': cannot unbind only meaningful in a <code>for',</code>while', or until' loop
- loop count
- %s %s %s
- OLDPWD=
- HOME not set
- OLDPWD not set
- CDPATH
- command_builtin
- line %d:
- %s: usage:
- %s: numeric argument required
- %s: not found
- %s: invalid option name
- %s: invalid number
- %s: invalid signal specification
- %s': not a pid or valid job spec %s: readonly variable argument %s: %s out of range %s out of range %s: no such job %s: no job control no job control %s: restricted restricted %s: not a shell builtin write error: %s %s: error retrieving current directory: %s: %s %s: ambiguous job spec SIGJUNK Unknown %2d) %s +afiprtxF cannot use -f' to make functions
- %s %d %s
- %s: cannot destroy array variables in this way
- can only be used in a function
- adnpsf:
- enable %s
- enable -n %s
- cannot open shared object %s: %s
- cannot find %s in shared object %s: %s
- %s: not dynamically loaded
- %s: cannot delete: %s
- %s: not a regular file
- %s: file is too large
- _evalfile
- source
- parseandexecute_top
- pe_dispose
- parseandexecute
- %s: cannot execute: %s
- ~/.bash_logout
- There are stopped jobs.
- not login shell: use exit' logout :e:lnrs no command found history specification bash-fc %s: cannot open temp file: %s ${FCEDIT:-${EDITOR:-ed}} ${FCEDIT:-${EDITOR:-vi}} fc builtin job %d started without job control builtin hash -p %s %s hashing disabled dlp:rt command %s: hash table empty These shell commands are defined internally. Type help' to see this list.
- Type help name' to find out more about the function name'.
- Use info bash' to find out more about the shell in general. Use <code>man -k' or</code>info' to find out more about commands not in this list. A star (*) next to a name means that the command is disabled. Shell commands matching keywords
- Shell commands matching keyword %s: %s %s: cannot open: %s no help topics match <code>%s'. Try</code>help help' or <code>man -k %s' or</code>info %s'. acd:npsrw cannot

use more than one of -anrw %s: history expansion failed history position %5d%c %s%s HISTFILE current no other options allowed with-x'

- lpnxrs
- jobs_builtin
- Unknown error
- (%ld) - %s
- %s: arguments must be process or job IDs
- directory stack empty
- directory stack index
- dirstack
-
- %2d %s
- %2d %s
- no other directory
- %s: invalid timeout specification
- %s: invalid file descriptor specification
- %d: invalid file descriptor: %s
- ersa:d:n:p:t:u:
- read_builtin
- read error: %d: %s
- can only return' from a function or sourced script allexport braceexpand errexit errtrace functrace hashall histexpand history interactive-comments monitor noclobber noexec noglob notify nounset onecmd physical pipefail privileged xtrace SHELLOPTS cannot simultaneously unset a function and a variable %s: cannot unset variable %s: cannot unset: readonly %s %s: not an array variable declare -%s %s -%s %s: not a function shift count filename argument required %s: file not found cannot suspend cannot suspend a login shell unknown trap -- %s %d trap -- %s %s %s is aliased to%s'
- alias %s=%s
- %s is a shell keyword
- %s is a function
- %s is a shell builtin
- %s is %s
- %s is hashed (%s)
- core file size
- blocks
- data seg size
- kbytes
- file size
- max locked memory
- max memory size
- open files
- pipe size
- 512 bytes
- stack size
- cpu time
- seconds
- max user processes
- virtual memory
- (%s, -%c)
- (-%c)
- %-20s %16s

- *unlimited*
- *%s: cannot get limit: %s*
- %c': bad command %s: cannot modify limit: %s u=%s,g=%s,o=%s *%c': invalid symbolic mode operator*
- %c': invalid symbolic mode character octal number umask%s OPTARG cdable_vars cdspell checkhash checkwinsize cmdhist dotglob execfail expand_aliases extdebug extglob extquote failglob force_fignore gnu_errfmt histappend histreedit histverify hostcomplete huponexit interactive_comments lithist login_shell mailwarn noemptycmd_completion nocaseglob nocasematch nullglob progcomp promptvars restricted_shell shift_verbose sourcepath xpg_echo %s: invalid shell option name shopt %s %s cannot set and unset shell options simultaneously set %co %s missing hex digit for \x warning: %s: %s #'-+ 0 hjlLtz *%s': missing format character*
- %c': invalid format character %s: illegal option -- %c %s: option requires an argument -- %c arrayvar binding directory disabled enabled export helptopic hostname keyword running service setopt signal stopped bashdefault default dirnames filenames nospace plusdirs %s: invalid action name abcdefgjko:prsuvA:G:W:P:S:X:F:C: complete -o %s -A %s %s %s warning: -F option may not work as you expect warning: -C option may not work as you expect compgen %s: no completion specification BASH_REMATCH /dev/fd/ /dev/std %s: bad network path specification socket connect %ldm%d.%03ds sh_makepath sh_realpath TMPDIR /var/tmp/ /var/tmp /usr/tmp %s/%s-%lu %s/cur %s/tmp %s/new invalid base 0123456789ABCDEF 0123456789abcdef hl^$0ftFT;,%wbeWBE| _*\AaIiCcDdPpYyRrSsXx~ l|h^0%bB l|h^0bB l|hwW^0bB accept-line arrow-key-prefix backward-byte backward-char backward-delete-char backward-kill-line backward-kill-word backward-word beginning-of-history beginning-of-line call-last-kbd-macro capitalize-word character-search character-search-backward clear-screen copy-backward-word copy-forward-word copy-region-as-kill delete-char delete-char-or-list delete-horizontal-space digit-argument do-lowercase-version downcase-word dump-functions dump-macros dump-variables emacs-editing-mode end-kbd-macro end-of-history end-of-line exchange-point-and-mark forward-backward-delete-char forward-byte forward-char forward-search-history forward-word history-search-backward history-search-forward insert-comment insert-completions kill-whole-line kill-line kill-region kill-word menu-complete next-history non-incremental-forward-search-history non-incremental-reverse-search-history non-incremental-forward-search-history-again non-incremental-reverse-search-history-again overwrite-mode possible-completions previous-history quoted-insert re-read-init-file redraw-current-line reverse-search-history revert-line self-insert set-mark start-kbd-macro tab-insert tilde-expand transpose-chars transpose-words tty-status universal-argument unix-filename-rubout unix-line-discard unix-word-rubout upcase-word yank-last-arg yank-nth-arg yank-pop vi-append-eol vi-append-mode vi-arg-digit vi-back-to-indent vi-bWord vi-bword vi-change-case vi-change-char vi-change-to vi-char-search vi-column vi-complete vi-delete vi-delete-to vi-eWord vi-editing-mode vi-end-word vi-eof-maybe vi-eword vi-fWord vi-fetch-history vi-first-print vi-fword vi-goto-mark vi-insert-beg vi-insertion-mode

` vi-match vi-movement-mode vi-next-word vi-overstrike vi-overstrike-delete vi-prev-word vi-put vi-redo vi-replace vi-rubout vi-search vi-search-again vi-set-mark vi-subst vi-tilde-expand vi-yank-arg vi-yank-to "\'` @$><=;|&{(

- *--More--*
- *Display all %d possibilities? (y or n)*
- *readline: bad value %d for whattodo in rl_complete*
- *Control-*
- *bell-style*
- *completion-query-items*
- *editing-mode*
- *isearch-terminators*
- *keymap*
- *bind-tty-special-chars*
- *blink-matching-paren*
- *byte-oriented*
- *completion-ignore-case*
- *convert-meta*
- *disable-completion*
- *enable-keypad*
- *expand-tilde*
- *history-preserve-point*
- *horizontal-scroll-mode*
- *input-meta*
- *mark-directories*
- *mark-modified-lines*
- *mark-symlinked-directories*
- *match-hidden-files*
- *meta-flag*
- *output-meta*
- *page-completions*
- *prefer-visible-bell*
- *print-completions-horizontally*
- *show-all-if-ambiguous*
- *show-all-if-unmodified*
- *visible-stats*
- *emacs-standard*
- *emacs-meta*
- *emacs-ctlx*
- *vi-move*
- *vi-command*
- *vi-insert*
- *include*
- *Escape*
- *Newline*
- *Return*
- *Rubout*
- *readline: %s: line %d: %s*
- *readline: %s*
- *$else found without matching $if*
- *$endif without matching $if*
- *audible*
- *visible*

- # %s (not bound)
- "%s": %s
- %s is not bound to any keys
- %s can be found on
- "%s"%s
- set %s %s
- %s is set to %s' "%s%s": "%s" %s%s outputs %s unknown parser directive no closing"' in key binding
- prefix-meta
- INPUTRC
- ~/.inputrc
- /etc/inputrc
- readline: debug: insertsomechars: count (%d) != col (%d)
- /-_=~.#$
- readline: readlinecallbackread_char() called with no handler!
- (arg: %d)
- ;&()|<>
- no previous substitution
- unknown expansion error
- event not found
- bad word specifier
- substitution failed
- unrecognized history modifier
- <>;&|$
- xdigit
- backspace
- vertical-tab
- form-feed
- carriage-return
- exclamation-mark
- quotation-mark
- number-sign
- dollar-sign
- percent-sign
- ampersand
- apostrophe
- left-parenthesis
- right-parenthesis
- asterisk
- plus-sign
- hyphen
- hyphen-minus
- period
- full-stop
- solidus
- semicolon
- less-than-sign
- equals-sign
- greater-than-sign
- question-mark
- commercial-at
- left-square-bracket
- backslash

- *reverse-solidus*
- *right-square-bracket*
- *circumflex*
- *circumflex-accent*
- *underscore*
- *grave-accent*
- *left-brace*
- *left-curly-bracket*
- *vertical-line*
- *right-brace*
- *right-curly-bracket*
- */locale.alias*
- */usr/share/locale*
- *messages*
- *LC_XXX*
- *LC_MONETARY*
- *LANGUAGE*
- *GETTEXTLOGUNTRANSLATED*
- *plural=*
- *nplurals=*
- *domain*
- *msgid*
- *msgid_plural*
- *msgstr[0] ""*
- *msgstr ""*
- *!"#$&'(),-/02479:<=>?@ABCEFGHIVWXYZ,OPU*
- *4 !"#$&()/0*
- *:9;,-S*
- *,QRXU+U*
- *359UUWWUYP*
- *2PP5>DUWW*+Y
- JUUUUUKLMU
- TU3PPOO
- 3UPUUU5
- 6777788999999999999999999999999999999:::;;<<====>>>>>>>>>>>????????@@@@AAAAAABBBCCCDDEFFFGHIJJJKKLLLLL
- -0:41/65789
- %+(.uv
- *zzzzzz*
- EF*zzzz*
- XopqrsOe zHQWzzzzfN zzCDIJ zhabzz zzz^cd !"#$%&'()- _mhexecute_header __DefaultRuneLocale _divdi3 _error _maskrune _mbcurmax _moddi3 _stderrp _stdinp _stdoutp _tolower _toupper _udivdi3 _umoddi3 _abort _access _alarm _asprintf _bcopy _bsearch _chdir _clearerr _close$UNIX2003 _closedir$UNIX2003 compatmode _confstr$UNIX2003 _connect$UNIX2003 _dlclose _dlerror _dlopen _dlsym _endgrent _endpwent _endservent _execve _fclose _fcntl$UNIX2003 _fdopen _ferror _fflush _fgets _fileno _fopen _fprintf _fputc _fputs$UNIX2003 _freeaddrinfo _fstat _fwrite$UNIX2003 gaistrerror _getaddrinfo _getcwd _getdtablesize

```
<li>_getegid</li> <li>_geteuid</li> <li>_getgid</li> <li>_getgrent</li> <li>_getgroups</li>
<li>_gethostname</li> <li>_getpeername$UNIX2003</li> <li>_getpgrp</li> <li>_getpid</li> <li>_getppid</li>
<li>_getpwent</li> <li>_getpwnam</li> <li>_getpwuid</li> <li>_getrlimit$UNIX2003</li> <li>_getrusage</li>
<li>_getservent</li> <li>_gettimeofday</li> <li>_getuid</li> <li>_ioctl</li> <li>_isatty</li>
<li>_kill$UNIX2003</li> <li>_killpg$UNIX2003</li> <li>_localtime</li> <li>_lseek</li> <li>_lstat</li>
<li>_malloc</li> <li>_mblen</li> <li>_mbrlen</li> <li>_mbrtowc</li> <li>_mbsinit</li> <li>_mbsrtowcs</li>
<li>_mbstowcs</li> <li>_mbtowc</li> <li>_memcmp</li> <li>_memcpy</li> <li>_memmove</li> <li>_memset</li>
<li>_mmap$UNIX2003</li> <li>_munmap$UNIX2003</li> <li>_open$UNIX2003</li> <li>_opendir$UNIX2003</li>
<li>_pathconf</li> <li>_printf</li> <li>_putchar</li> <li>_qsort</li> <li>_raise</li>
<li>_read$UNIX2003</li> <li>_readdir</li> <li>_readlink</li> <li>_realloc</li> <li>_regcomp$UNIX2003</li>
<li>_regexec</li> <li>_regfree</li> <li>_select$UNIX2003</li> <li>_setgid</li> <li>_setgrent</li>
<li>_setlocale</li> <li>_setpgid</li> <li>_setpwent</li> <li>_setregid$UNIX2003</li>
<li>_setreuid$UNIX2003</li> <li>_setrlimit$UNIX2003</li> <li>_setservent</li> <li>_setuid</li>
<li>_setvbuf</li> <li>_sigaction</li> <li>_sigaddset</li> <li>_sigemptyset</li> <li>_siglongjmp</li>
<li>_sigprocmask</li> <li>_sigsetjmp</li> <li>_sleep$UNIX2003</li> <li>_snprintf</li> <li>_socket</li>
<li>_sprintf</li> <li>_stpcpy</li> <li>_strcasecmp</li> <li>_strcat</li> <li>_strchr</li> <li>_strcmp</li>
<li>_strcoll</li> <li>_strcpy</li> <li>_strdup</li> <li>_strerror$UNIX2003</li> <li>_strftime$UNIX2003</li>
<li>_strlen</li> <li>_strncasecmp</li> <li>_strncmp</li> <li>_strncpy</li> <li>_strpbrk</li>
<li>_strrchr</li> <li>_strsignal</li> <li>_strstr</li> <li>_strtod$UNIX2003</li> <li>_strtoimax</li>
<li>_strtoul</li> <li>_strtoumax</li> <li>_sysconf</li> <li>_tcgetattr</li> <li>_tcgetpgrp</li>
<li>_tcsetattr</li> <li>_tcsetpgrp</li> <li>_tgetent</li> <li>_tgetflag</li> <li>_tgetnum</li>
<li>_tgetstr</li> <li>_tgoto</li> <li>_tputs</li> <li>_ttyname</li> <li>_tzset</li> <li>_umask</li>
<li>_unlink</li> <li>_vfprintf</li> <li>_vsnprintf</li> <li>_waitpid$UNIX2003</li> <li>_wcrtomb</li>
<li>_wcschr</li> <li>_wcscmp</li> <li>_wcscoll</li> <li>_wcscpy</li> <li>_wcslen</li> <li>_wcsncmp</li>
<li>_wcsrtombs</li> <li>_wctob</li> <li>_wctype</li> <li>_wcwidth</li> <li>_write$UNIX2003</li>
<li>com.apple.bash</li> <li>.dkv'n</li> <li>ZM$fW,</li> <li>P}wBB?s</li> <li>Apple Inc.1&amp;0$</li>
<li>Apple Certification Authority1</li> <li>Apple Root CA0</li> <li>070214211919Z</li>
<li>150214211919Z0</li> <li>Apple Inc.1&amp;0$</li> <li>Apple Certification Authority1301</li> <li>*Apple
Code Signing Certification Authority0</li> <li>drm|iGa</li> <li>%http://www.apple.com/appleca/root.crl0</li>
<li>QKK:IaN</li> <li>Apple Inc.1&amp;0$</li> <li>Apple Certification Authority1</li> <li>Apple Root CA0</li>
<li>060425214036Z</li> <li>350209214036Z0b1</li> <li>Apple Inc.1&amp;0$</li> <li>Apple Certification
Authority1</li> <li>Apple Root CA0</li> <li>https://www.apple.com/appleca/0</li> <li>Reliance on this
certificate by any party assumes acceptance of the then applicable standard terms a</li> <li>Apple
Inc.1&amp;0$</li> <li>Apple Certification Authority1301</li> <li>*Apple Code Signing Certification
Authority0</li> <li>070223220256Z</li> <li>150114220256Z0V1</li> <li>Apple Inc.1</li> <li>Apple
Software1</li> <li>Software Signing0</li> <li>EA3%|k=</li> <li>http://www.apple.com/appleca/0</li>
<li>Reliance on this certificate by any party assumes acceptance of the then applicable standard terms
a</li> <li>,http://www.apple.com/appleca/codesigning.crl0</li> <li>Apple Inc.1&amp;0$</li> <li>Apple
Certification Authority1301</li> <li>*Apple Code Signing Certification Authority</li> <li>8__PAGEZERO</li>
<li>__TEXT</li> <li>__text</li> <li>__TEXT</li> <li><em>_symbol</em>stub1</li> <li>__TEXT</li>
<li>__cstring</li> <li>__TEXT</li> <li>__const</li> <li>__TEXT</li> <li>__literal8</li> <li>__TEXT</li>
<li>__DATA</li> <li>__data</li> <li>__DATA</li> <li>__dyld</li> <li>__DATA</li> <li>
<em>_la</em>symbol_ptr</li> <li>__DATA</li> <li><em>_nl</em>symbol_ptr</li> <li>__DATA</li> <li>__const</li>
<li>__DATA</li> <li>__DATA</li> <li>__common</li> <li>__DATA</li> <li>8__LINKEDIT</li>
<li>/usr/lib/dyld</li> <li>/usr/lib/libncurses.5.4.dylib</li> <li>/usr/lib/libiconv.2.dylib</li>
<li>/usr/lib/libSystem.B.dylib</li> <li>/usr/lib/libgcc_s.1.dylib</li> <li>:8B.p|</li> <li>9ka&lt;= </li>
<li>x8cbxH</li> <li>:8Bcx|</li> <li>D8cdPH</li> <li>0}c[xH</li> <li>L8cMl8</li> <li>&lt;}#KxK</li>
<li>D8cf0H</li> <li>D8cflH</li> <li>D8cglH</li> <li>8Bd:}+</li> <li>9Jd:|]J</li> <li>}j[x|g</li>
<li>x8chpH</li> <li>:|BH.H</li> <li>:|bH.H</li> <li>:|bH.H</li> <li>:|bH.H</li> <li>8cjL8!</li>
<li>x8cj\K</li> <li>x8ck0K</li> <li>8ckt8B</li> <li>=@ff9
```


- H|c6p|c
- TTc02;
- }9Kx}^Sx

- }#Kx;
- d8qm8?
- 8B09TC
- }c[xH
- |iY.TJ
- :|bH.H
- :|bH.K
- :|bH.K
- }#KxH
- }+Kxa)
- x;9pl;
- @<@ff}K
- 8})PP9)
- |B.p|@
- D8cr(K
- 8B},9)|
- 8B~ 9)~ |/|BH0|I 8caL8! :|bP.H x8ct H
- |BX0|I
- @}#Kx}DSx
- }^Sx}=Kx
- }^Sx}=Kx
- 8}>Kx}*Sx*
- *"x}>Kx}*Sx
- #x}>Kx}*Sx*
- *l}c[xH*
- *vH|JX.*
- *x9)vt9*
- *,|cFp/*
- *D|@Fp/*
- *:|B@./*
- *0}c[x8*
- *} FpT*
- *l} FpT*
- *|@Fp8@*
- *0}"Fp/*
- *h} Fp/*
- |XP8B
- }BY.9k
- } Fp8@ 8}@Fp/ |@Fp/ |BFp;b d} Fp/ p}@FpT } Rx @
- }@Zx @
- } Rx @ } Rx @
- :scP;@
- :scL:RcP;@
- Cx}:Kx|~
- }$Kx}#
- x}e[xH
- xTB80;b
- x}+KxH
- }{[x}+Kx/
- x8c{,H
- }i[x<@
- }i[x<@
- }

- }i[x<@
- }$Kx}#
- };Kx}:
- <} cyA 8re$~ }?X.<@ 0}#KxK }c[x8! |_HP8B :}iX.A |_XP8B :||H./ :8B^|| :8BpT| :8BtP| :8Bx@| 8|bP.K :8Bzd| }c[x|
- }c[x|
- +x}c[x8
- +x}c[x8
- +x}c[x8
- +x}c[x8
- +x}c[x8
- +x}c[x8
- +x}c[x8
- +x}c[x8
- +x}c[x8
- }c[xH
- .}i[xH
- x}ESx.
- }ESx}D
- xTB 6;"
- }"KxH
- x|d(P|
- |BH0|I
- |BH0|I
- x},KxN
- }DSx<@
- }DSx<@
- }#KxH
- PTi(4Tc
- x8cbxN
- 88c:PH
- | #y|
- x}#Kx}"
- :|bH.H
- :|bH.K
- :|bH.H
- :|bH.K
- :|bH.H
- H.|bI.H
- :RA|:sA
- ,<;Z,HH
- x8|9dK
- x8cBHH
- ;9/x;ZB
- d|JSxT
- @8cDHH
- }lSx< x8cGTH :8Byt| }h[x9k ID8cI< IH8cI< IL8cI< IP8cI< IT8cI< Tc 68c P}bXP/ }jHP|K p|B6p|I P}knp})np| }eXP}$HP} PP|f@P P}?Kx/ x|bH.N }]Sx8@ }]Sx9k >}?KxH UJ 68K UJ 68K =@ffaJfg| |BP8B x8cK4H x8cK8H x8cK<H x8cK@H x8cKDH x8cKHH L}c[xN ,}#KxH x}i[x| L}<code>[x9</code> }$Kx8

- }$Kx8 }$Kx|E }$Kx|E P}c[xH x},KxN ;ZjD<@ }*Kx9) :|bH.H 4|bH.H |]PP|}J ,}"Kx9
- }LSx9)
- }5KxB@
- X}#KxK
- P|BVp|B
- L|BVp|B
- }cPP<@
- :|bH.H
- } HP9)
- }l[x?@
- D9)Z ?@
- }d[x|iX
- }c[x}DSxK
- 0}CSx}d[xK
- }CSx}d[xK
- :|bH.N
- }i[x|d
- x}x[xH
- x||hP8c
- @}: P|
- @|Z P8B
- P}&Kx|
- <}*KxH*
- *<}*KxH
- }'Kx8@
- x}$KxH
- x}$KxH
- x}$KxH
- x}$KxH
- x}$KxH
- x}$KxH
- x}$KxH
- F>|BKx|B[x|J
- F>|BKx|B[x|B
- F>|BKx|B[x|B
- F>|BKx|B[x|B
- F>|BKx|B[x|@
- F>|BKx|B[x|B
- F>|BKx|B[x|M
- })Cx}kSxT
- F>TBF>})Cx}kSx}2
- F>|BKx|B[x|@
- F>|BKx|B[x|B
- F>|BKx|B[x|B
- F>|BKx|B[x|@
- .U)F>|
- .UlF>|
- F>|BKx|B[x|B
- .UlF>|
- .U)F>|
- F>|BKx|B[x|@
- .TBF>|

- F>|BKx|B[x|@
- })Cx}kSxT
- F>TBF>})Cx}kSx}?
- x})[x})
- 0}"P.}/Q.9J
- }@@P|BY
- :|OX./
- TD 6T]
- Cx}/Kx}NSx
- 8pB33|
- }9Kx}XSx
- $U@ 69)
- T@F>}@
- |BPP;"
- x})[x}"
- F>|BKx|B[x|B
- x})[x}"
- x})[x})
- x})[x}#
- F>|BKx|B[x|B
- ; N `B |}x[x8 :8BE8| <8B~D| :8BI@| 8B}h9) <8B~D| <8B~D| debug debugger dump-po-strings dump-strings init-file noediting noprofile protected rcfile restricted verbose version wordexp ~/.bashrc BASH_ENV run_wordexp --wordexp runonecommand /bin/sh I have no name! ??host?? GNU bash, version %s-(%s) powerpc-apple-darwin9.0 Usage: %s [GNU long option] [option] ... %s [GNU long option] [option] script-file ... GNU long options: Shell options: -irsD or -c command or -O shopt_option (invocation only) -%s or -o option Type` %s -c "help set"' for more information about shell options.
- Type `%s -c help' for more information about shell builtin commands.</li> <li>Use the` bashbug' command to report bugs.
- %s: option requires an argument
- %s: invalid option
- %c%c: invalid option
- bin/sh
- Unix2003
- POSIXLY*CORRECT*
- *POSIX*PEDANTIC
- \s-\v\$
- *BASH*IMPLICIT*DASH*PEE
- setuid(%u) failed: %s
- setreuid(%u,%u) failed: %s
- /etc/profile
- ~/.profile
- ~/.bash*profile*
- *~/.bash*login
- /usr/local/share/bashdb/bashdb-main.inc
- FUNCNAME
- BASH*SOURCE*
- *BASH*LINENO
- %s: is a directory
- %s: cannot execute binary file
- timed out waiting for input: auto-logout

- PROMPT*COMMAND*
- *reader*loop
- newline
- select
- function
- %a %b %d
- %H:%M:%S
- %I:%M:%S
- %I:%M %p
- syntax error near unexpected token %s' syntax error near %s'
- syntax error: unexpected end of file
- syntax error
- unexpected EOF while looking for matching %c' readline stdin PIPESTATUS unexpected EOF while looking for]]'
- syntax error in conditional expression: unexpected token %s' syntax error in conditional expression unexpected token <code>%s', expected</code>)' expected)'
- unexpected argument %s' to conditional unary operator unexpected argument to conditional unary operator unexpected token %s', conditional binary operator expected
- conditional binary operator expected
- unexpected argument %s' to conditional binary operator unexpected argument to conditional binary operator unexpected token %c' in conditional command
- unexpected token %s' in conditional command unexpected token %d in conditional command unexpected EOF while looking for matching)'
- Use "%s" to leave the shell.
- logout
- memory exhausted
- getcwd: cannot access parent directories
- /dev/tty
- OLDPWD
- %s': not a valid identifier makeheredocument: bad instruction type %d make_redirection: redirection instruction %d' out of range
- clean*simple*command
- syntax error: arithmetic expression required
- syntax error: ;' unexpected syntax error: ((%s))'
- %s=(%s)
- for %s in
- select %s in
- case %s in
- cprintf: %c': invalid format character <<%s%s <<< %s %d<&%d %d>&%d %d<&%s %d>&%s %d<&%d- %d>&%d- %d<&%s- %d>&%s- case %s in for ((print_command: bad connector %d'
- function %s ()
- print*command*
- *dispose*command
- /dev/null
- cannot redirect standard input from /dev/null: %s
- execute*cond*node
- eval*builtin*
- *builtin*env
- return*temp*env
- %s: %s: bad interpreter
- cannot duplicate fd %d to fd %d

- real %2R
- user %2U
- sys %2S
- TIMEFORMAT
- TIMEFORMAT: %c': invalid format character loop_redirections %s: readonly function execute_command function_calling execute-shell-function simple-command auto_resume substring saved redirects saved_redirects %s: restricted: cannot specify/' in command names
- %s: command not found
- execute-command
- COLUMNS
- *%d%s%s*
- *pipe error*
- *pipe-file-descriptors*
- *execute_connection*
- *COMP_WORDBREAKS*
- *GLOBIGNORE*
- *HISTCONTROL*
- *HISTFILESIZE*
- *HISTIGNORE*
- *HISTSIZE*
- *HISTTIMEFORMAT*
- *HOSTFILE*
- *IGNOREEOF*
- *LC_ALL*
- *LC_COLLATE*
- *LC_CTYPE*
- *LC_MESSAGES*
- *LC_NUMERIC*
- *LC_TIME*
- *MAILCHECK*
- *MAILPATH*
- *OPTERR*
- *OPTIND*
- *TERMCAP*
- *TERMINFO*
- *TEXTDOMAIN*
- *TEXTDOMAINDIR*
- *histchars*
- *ignoreeof*
- *alllocalvariables: no function context at current scope*
- *popvarcontext: head of shell_variables not a function context*
- *popvarcontext: no global_variables context*
- *shell-init*
- *ignorespace*
- *ignoredups*
- *ignoreboth*
- *erasedups*
- *HISTFILE*
- *shell level (%d) too high, resetting to 1*
- *makelocalvariable: no function context at current scope*
- *BASH_ARGV*
- *BASH_ARGC*

- popscope: head of shellvariables not a temporary environment scope
- error importing function definition for %s' /usr/gnu/bin:/usr/local/bin:/bin:/usr/bin:. HOSTTYPE powerpc OSTYPE darwin9.0 MACHTYPE HOSTNAME BASH_VERSION BASH_VERSINFO BASHEXECUTIONSTRING ~/.sh_history ~/.bash_history SECONDS BASH_COMMAND BASH_SUBSHELL RANDOM LINENO HISTCMD DIRSTACK GROUPS bash-maintainers@gnu.org unknown command error bad command type bad connector bad jump line %s:%s%d: last command: %s (null) Aborting... %s: %s %s: warning: %s: %s:%s%d: %s: %s: %d %s: unbound variable %s: readonly variable %s%s%s: %s (error token is "%s") invalid number invalid arithmetic base value too great for base */%+-&^| syntax error: operand expected syntax error: invalid arithmetic operator identifier expected after pre-increment or pre-decrement missing)'
- exponent less than 0
- division by 0
- expression expected
- :' expected for conditional expression attempted assignment to non-variable bug: bad expassign token expression recursion level exceeded syntax error in expression recursion stack underflow <unknown> [%d] %ld describe_pid: %ld: no such pid Signal %d [%d]%c Stopped(%s) Stopped Running Done(%d) Exit %d Unknown status (core dumped) (wd: %s) [%ld: %d] tcsetattr %s: line %d: (core dumped) (wd now: %s) notifyofjob_status initializejobcontrol: getpgrp failed initializejobcontrol: setpgid no job control in this shell deleting stopped job %d with process group %ld SIGCHLD trap job-working-directory wait_for: No record of process %ld %s: job has terminated %s: job %d already in background [%d]%s (wd: %s) waitforjob: job %d is stopped wait: pid %ld is not a child of this shell child setpgid (%ld to %ld) forked pid %d appears in running job %d cannot make pipe for command substitution cannot make child for command substitution command_substitute: cannot duplicate pipe as fd 1 command substitution %s: cannot assign list to array member bad substitution: no closing %s' in %s
- cannot make pipe for process substitution
- /dev/fd/
- cannot make child for process substitution
- cannot duplicate named pipe %s as fd %d
- process substitution
- bad substitution: no closing " " in %s no match: %s %s: parameter null or not set #%:-=?+/} %s: substring expression < 0 %s: bad substitution $%s: cannot assign in this way /var/mail You have mail in $_ You have new mail in $_ The mail in %s has been read SIGHUP SIGINT SIGQUIT SIGILL SIGTRAP SIGABRT SIGEMT SIGFPE SIGKILL SIGBUS SIGSEGV SIGSYS SIGPIPE SIGALRM SIGTERM SIGURG SIGSTOP SIGTSTP SIGCONT SIGCHLD SIGTTIN SIGTTOU SIGXCPU SIGXFSZ SIGVTALRM SIGPROF SIGWINCH SIGINFO SIGUSR1 SIGUSR2 RETURN invalid signal number exit trap interrupt trap debug trap error trap return trap runpendingtraps: bad value in trap_list[%d]: %p runpendingtraps: signal handler is SIG_DFL, resending %d (%s) to myself trap_handler: bad signal %d cannot allocate new file descriptor for bash input from fd %d savebashinput: buffer already exists for new fd %d argument expected %s: integer expression expected %s: unary operator expected %s: binary operator expected)' expected
-)' expected, found %s missing]'
- too many arguments

- @(#)Bash version 3.2.17(1) release GNU
- release
- %s.%d(%d)-%s
- %s.%d(%d)
- GNU bash, version %s (%s)
- Copyright (C) 2005 Free Software Foundation, Inc.
- bad array subscript
- array assign
- %s[%s: %s
- [%s]: %s
- %s: cannot assign to non-numeric index
- ;&()|<>
- %s: cannot create: %s
- %s%s%s
- FIGNORE
- '"><=;|&(:
- '"@><=;|&(:
- comment-begin
- shell-expand-line
- history-expand-line
- magic-space
- alias-expand-line
- history-and-alias-expand-line
- insert-last-argument
- operate-and-get-next
- display-shell-version
- edit-and-execute-command
- complete-into-braces
- complete-filename
- possible-filename-completions
- complete-username
- possible-username-completions
- complete-hostname
- possible-hostname-completions
- complete-variable
- possible-variable-completions
- complete-command
- possible-command-completions
- glob-complete-word
- glob-expand-word
- glob-list-expansions
- dynamic-complete-history
- \'"@<>=;|&()#$?*[!:{ $include hostnamecompletionfile /etc/hosts C-xC-e fc -e vi fc -e "${VISUAL:-${EDITOR:-vi}}" fc -e "${VISUAL:-${EDITOR:-emacs}}" completion-ignore-case ;|&{(
- ()<>;&|
- symlink-hook
- bashexecuteunix_command: cannot find keymap for command
- bashexecuteunix_command
- %s: first non-whitespace character is not "' no closing%c' in %s
- %s: missing colon separator
- /usr/share/locale
- #: %s:%d

- *msgid %s%s*
- *msgstr ""*
- */dev/tcp//*
- */dev/udp//*
- *file descriptor out of range*
- *%s: ambiguous redirect*
- *%s: cannot overwrite existing file*
- *%s: restricted: cannot redirect output*
- *cannot create temp file for here document: %s*
- *redirection error: cannot duplicate fd*
- *sh-thd*
- *COMP_LINE*
- *COMP_POINT*
- *COMP_WORDS*
- *COMP_CWORD*
- *completion: function* %s' not found COMPREPLY progcomp_insert: %s: NULL COMPSPEC xmalloc: cannot allocate %lu bytes (%lu bytes allocated) xrealloc: cannot reallocate %lu bytes (%lu bytes allocated) Display the possible completions depending on the options. Intended to be used from within a shell function generating possible completions. If the optional WORD argument is supplied, matches against WORD are generated. For each NAME, specify how arguments are to be completed. If the -p option is supplied, or if no options are supplied, existing completion specifications are printed in a way that allows them to be reused as input. The -r option removes a completion specification for each NAME, or, if no NAMEs are supplied, all completion specifications. printf formats and prints ARGUMENTS under control of the FORMAT. FORMAT is a character string which contains three types of objects: plain characters, which are simply copied to standard output, character escape sequences which are converted and copied to the standard output, and format specifications, each of which causes printing of the next successive argument. In addition to the standard printf(1) formats, %b means to expand backslash escape sequences in the corresponding argument, and %q means to quote the argument in a way that can be reused as shell input. If the -v option is supplied, the output is placed into the value of the shell variable VAR rather than being sent to the standard output. Toggle the values of variables controlling optional behavior. The -s flag means to enable (set) each OPTNAME; the -u flag unsets each OPTNAME. The -q flag suppresses output; the exit status indicates whether each OPTNAME is set or unset. The -o option restricts the OPTNAMEs to those defined for use with *set -o'. With no options, or with the -p option, a list of all*
- *settable options is displayed, with an indication of whether or*
- *not each is set.*
- *Display the list of currently remembered directories. Directories*
- *find their way onto the list with the* pushd' command; you can get back up through the list with the*popd' command.*
- *The -l flag specifies that* dirs' should not print shorthand versions of directories which are relative to your home directory. This means that <code>~/bin' might be displayed as</code>/homes/bfox/bin'. The -v flag causes*dirs' to print the directory stack with one entry per line,*
- *prepending the directory name with its position in the stack. The -p*
- *flag does the same thing, but the stack position is not prepended.*
- *The -c flag clears the directory stack by deleting all of the elements.*
- *+N*
- *displays the Nth entry counting from the left of the list shown by*
- *dirs when invoked without options, starting with zero.*
- *-N*
- *displays the Nth entry counting from the right of the list shown by*
- *dirs when invoked without options, starting with zero.*
- *Removes entries from the directory stack. With no arguments,*

- *removes the top directory from the stack, and cd's to the new*
- *top directory.*
- *+N*
- *removes the Nth entry counting from the left of the list*
- *shown by* dirs', starting with zero. For example: *popd +0'*
- *removes the first directory,* popd +1' the second. -N removes the Nth entry counting from the right of the list shown by <code>dirs', starting with zero. For example:</code>popd -0' removes the last directory, *popd -1' the next to last.*
- *-n*
- *suppress the normal change of directory when removing directories*
- *from the stack, so only the stack is manipulated.*
- *You can see the directory stack with the* dirs' command. Adds a directory to the top of the directory stack, or rotates the stack, making the new top of the stack the current working directory. With no arguments, exchanges the top two directories. +N Rotates the stack so that the Nth directory (counting from the left of the list shown by *dirs', starting with*
- *zero) is at the top.*
- *-N*
- *Rotates the stack so that the Nth directory (counting*
- *from the right of the list shown by* dirs', starting with zero) is at the top. -n suppress the normal change of directory when adding directories to the stack, so only the stack is manipulated. dir adds DIR to the directory stack at the top, making it the new current working directory. You can see the directory stack with the *dirs' command.*
- *BASH_VERSION*
- *Version information for this Bash.*
- *CDPATH*
- *A colon-separated list of directories to search*
- *for directries given as arguments to* cd'. GLOBIGNORE A colon-separated list of patterns describing filenames to be ignored by pathname expansion. HISTFILE The name of the file where your command history is stored. HISTFILESIZE The maximum number of lines this file can contain. HISTSIZE The maximum number of history lines that a running shell can access. HOME The complete pathname to your login directory. HOSTNAME The name of the current host. HOSTTYPE The type of CPU this version of Bash is running under. IGNOREEOF Controls the action of the shell on receipt of an EOF character as the sole input. If set, then the value of it is the number of EOF characters that can be seen in a row on an empty line before the shell will exit (default 10). When unset, EOF signifies the end of input. MACHTYPE A string describing the current system Bash is running on. MAILCHECK How often, in seconds, Bash checks for new mail. MAILPATH A colon-separated list of filenames which Bash checks for new mail. OSTYPE The version of Unix this version of Bash is running on. PATH A colon-separated list of directories to search when looking for commands. PROMPT_COMMAND A command to be executed before the printing of each primary prompt. PS1 The primary prompt string. PS2 The secondary prompt string. PWD The full pathname of the current directory. SHELLOPTS A colon-separated list of enabled shell options. TERM The name of the current terminal type. TIMEFORMAT The output format for timing statistics displayed by the *time' reserved word.*
- *auto_resume*
- *Non-null means a command word appearing on a line by*
- *itself is first looked for in the list of currently*
- *stopped jobs. If found there, that job is foregrounded.*
- *A value of* exact' means that the command word must exactly match a command in the list of stopped jobs. A value of *substring' means that the command word must*
- *match a substring of the job. Any other value means that*
- *the command must be a prefix of a stopped job.*

- histchars
- Characters controlling history expansion and quick
- substitution. The first character is the history
- substitution character, usually !'. The second is the <code>quick substitution' character, usually</code>^'. The third is the <code>history comment' character, usually</code>#'. HISTIGNORE A colon-separated list of patterns used to decide which commands should be saved on the history list. Returns a status of 0 or 1 depending on the evaluation of the conditional expression EXPRESSION. Expressions are composed of the same primaries used by the test' builtin, and may be combined using the following operators
- (EXPRESSION)
- Returns the value of EXPRESSION
- ! EXPRESSION
- True if EXPRESSION is false; else false
- EXPR1 && EXPR2
- True if both EXPR1 and EXPR2 are true; else false
- EXPR1 || EXPR2
- True if either EXPR1 or EXPR2 is true; else false
- When the ==' and !=' operators are used, the string to the right of the
- operator is used as a pattern and pattern matching is performed. The
- && and || operators do not evaluate EXPR2 if EXPR1 is sufficient to
- determine the expression's value.
- The EXPRESSION is evaluated according to the rules for arithmetic
- evaluation. Equivalent to "let EXPRESSION".
- Equivalent to the JOB_SPEC argument to the fg' command. Resume a stopped or background job. JOB_SPEC can specify either a job name or a job number. Following JOB_SPEC with a &' places the job in
- the background, as if the job specification had been supplied as an
- argument to bg'. Run a set of commands in a group. This is one way to redirect an entire set of commands. Create a simple command invoked by NAME which runs COMMANDS. Arguments on the command line along with NAME are passed to the function as $0 .. $n. Expand and execute COMMANDS as long as the final command in the until' COMMANDS has an exit status which is not zero.
- Expand and execute COMMANDS as long as the final command in the
- while' COMMANDS has an exit status of zero. The if COMMANDS' list is executed. If its exit status is zero, then the
- then COMMANDS' list is executed. Otherwise, each elif COMMANDS' list is
- executed in turn, and if its exit status is zero, the corresponding
- then COMMANDS' list is executed and the if command completes. Otherwise, the else COMMANDS' list is executed, if present. The exit status of the
- entire construct is the exit status of the last command executed, or zero
- if no condition tested true.
- Selectively execute COMMANDS based upon WORD matching PATTERN. The
- |' is used to separate multiple patterns. Execute PIPELINE and print a summary of the real time, user CPU time, and system CPU time spent executing PIPELINE when it terminates. The return status is the return status of PIPELINE. The -p' option
- prints the timing summary in a slightly different format. This uses
- the value of the TIMEFORMAT variable as the output format.
- The WORDS are expanded, generating a list of words. The
- set of expanded words is printed on the standard error, each
- preceded by a number. If in WORDS' is not present, in "$@"'
- is assumed. The PS3 prompt is then displayed and a line read
- from the standard input. If the line consists of the number
- corresponding to one of the displayed words, then NAME is set
- to that word. If the line is empty, WORDS and the prompt are

- *redisplayed. If EOF is read, the command completes. Any other*
- *value read causes NAME to be set to null. The line read is saved*
- *in the variable REPLY. COMMANDS are executed after each selection*
- *until a break command is executed.*
- *Equivalent to*
- *((EXP1))*
- *while ((EXP2)); do*
- *COMMANDS*
- *((EXP3))*
- *EXP1, EXP2, and EXP3 are arithmetic expressions. If any expression is*
- *omitted, it behaves as if it evaluates to 1.*
- *The* for' loop executes a sequence of commands for each member in a list of items. If <code>in WORDS ...;' is not present, then</code>in "$@"' is assumed. For each element in WORDS, NAME is set to that element, and the COMMANDS are executed. Wait for the specified process and report its termination status. If N is not given, all currently active child processes are waited for, and the return code is zero. N may be a process ID or a job specification; if a job spec is given, all processes in the job's pipeline are waited for. The user file-creation mask is set to MODE. If MODE is omitted, or if <code>-S' is supplied, the current value of the mask is printed. The</code>-S' option makes the output symbolic; otherwise an octal number is output. If *-p' is supplied, and MODE is omitted, the output is in a form*
- *that may be used as input. If MODE begins with a digit, it is*
- *interpreted as an octal number, otherwise it is a symbolic mode string*
- *like that accepted by chmod(1).*
- *Ulimit provides control over the resources available to processes*
- *started by the shell, on systems that allowsuch control. If an*
- *option is given, it is interpreted as follows:*
- *-S*
- *use the* soft' resource limit -H use the *hard' resource limit*
- *-a*
- *all current limits are reported*
- *-c*
- *the maximum size of core files created*
- *-d*
- *the maximum size of a process's data segment*
- *-e*
- *the maximum scheduling priority (* nice') -f the maximum size of files written by the shell and its children -i the maximum number of pending signals -l the maximum size a process may lock into memory -m the maximum resident set size -n the maximum number of open file descriptors -p the pipe buffer size -q the maximum number of bytes in POSIX message queues -r the maximum real-time scheduling priority -s the maximum stack size -t the maximum amount of cpu time in seconds -u the maximum number of user processes -v the size of virtual memory -x the maximum number of file locks If LIMIT is given, it is the new value of the specified resource; the special LIMIT values <code>soft',</code>hard', and *unlimited' stand for*
- *the current soft limit, the current hard limit, and no limit, respectively.*
- *Otherwise, the current value of the specified resource is printed.*
- *If no option is given, then -f is assumed. Values are in 1024-byte*
- *increments, except for -t, which is in seconds, -p, which is in*
- *increments of 512 bytes, and -u, which is an unscaled number of*
- *processes.*
- *For each NAME, indicate howit would be interpreted if used as a*
- *command name.*

- *If the -t option is used,* type' outputs a single word which is one of <code>alias',</code>keyword', <code>function',</code>builtin', <code>file' or</code>', if NAME is an alias, shell reserved word, shell function, shell builtin, disk file, or unfound, respectively. If the -p flag is used*, type' either returns the name of the disk*
- *file that would be executed, or nothing if* type -t NAME' would not return*file'.*
- *If the -a flag is used,* type' displays all of the places that contain an executable named*file'. This includes aliases, builtins, and*
- *functions, if and only if the -p flag is not also used.*
- *The -f flag suppresses shell function lookup.*
- *The -P flag forces a PATH search for each NAME, even if it is an alias,*
- *builtin, or function, and returns the name of the disk file that would*
- *be executed.*
- *The command ARG is to be read and executed when the shell receives*
- *signal(s) SIGNALSPEC. If ARG is absent (and a single SIGNALSPEC*
- *is supplied) or* -', each specified signal is reset to its original value. If ARG is the null string each SIGNAL_SPEC is ignored by the shell and by the commands it invokes. If a SIGNAL_SPEC is EXIT (0) the command ARG is executed on exit from the shell. If a SIGNAL_SPEC is DEBUG, ARG is executed after every simple command. If the*-p' option*
- *is supplied then the trap commands associated with each SIGNAL_SPEC are*
- *displayed. If no arguments are supplied or if only* -p' is given, trap prints the list of commands associated with each signal. Each SIGNAL_SPEC is either a signal name in <signal.h> or a signal number. Signal names are case insensitive and the SIG prefix is optional*. trap -l' prints*
- *a list of signal names and their corresponding numbers. Note that a*
- *signal can be sent to the shell with "kill -signal $$".*
- *Print the accumulated user and system times for processes run from*
- *the shell.*
- *This is a synonym for the "test" builtin, but the last*
- *argument must be a literal*]', to match the opening*['.*
- *Exits with a status of 0 (true) or 1 (false) depending on*
- *the evaluation of EXPR. Expressions may be unary or binary. Unary*
- *expressions are often used to examine the status of a file. There*
- *are string operators as well, and numeric comparison operators.*
- *File operators:*
- *-a FILE True if file exists.*
- *-b FILE True if file is block special.*
- *-c FILE True if file is character special.*
- *-d FILE True if file is a directory.*
- *-e FILE True if file exists.*
- *-f FILE True if file exists and is a regular file.*
- *-g FILE True if file is set-group-id.*
- *-h FILE True if file is a symbolic link.*
- *-L FILE True if file is a symbolic link.*
- *-k FILE True if file has its* sticky' bit set. -p FILE True if file is a named pipe. -r FILE True if file is readable by you. -s FILE True if file exists and is not empty. -S FILE True if file is a socket. -t FD True if FD is opened on a terminal. -u FILE True if the file is set-user-id. -w FILE True if the file is writable by you. -x FILE True if the file is executable by you. -O FILE True if the file is effectively owned by you. -G FILE True if the file is effectively owned by your group. -N FILE True if the file has been modified since it was last read. FILE1 -nt FILE2 True if file1 is newer than file2 (according to modification date). FILE1 -ot FILE2 True if file1 is older than file2. FILE1 -ef FILE2 True if file1 is a hard link to file2. String operators: -z STRING True if string is empty. -n STRING STRING True if string is not empty. STRING1 = STRING2 True if the strings are equal. STRING1 != STRING2 True if the strings are not equal.

 STRING1 < STRING2 True if STRING1 sorts before STRING2 lexicographically. STRING1 > STRING2 True if STRING1 sorts after STRING2 lexicographically. Other operators: -o OPTION True if the shell option OPTION is enabled. ! EXPR True if expr is false. EXPR1 -a EXPR2 True if both expr1 AND expr2 are true. EXPR1 -o EXPR2 True if either expr1 OR expr2 is true. arg1 OP arg2 Arithmetic tests. OP is one of -eq, -ne, -lt, -le, -gt, or -ge. Arithmetic binary operators return true if ARG1 is equal, not-equal, less-than, less-than-or-equal, greater-than, or greater-than-or-equal than ARG2. Suspend the execution of this shell until it receives a SIGCONT signal. The -f' if specified says not to complain about this

- being a login shell if it is; just suspend anyway.
- Read and execute commands from FILENAME and return. The pathnames
- in $PATH are used to find the directory containing FILENAME. If any
- ARGUMENTS are supplied, they become the positional parameters when
- FILENAME is executed.
- The positional parameters from $N+1 ... are renamed to $1 ... If N is
- not given, it is assumed to be 1.
- The given NAMEs are marked readonly and the values of these NAMEs may
- not be changed by subsequent assignment. If the -f option is given,
- then functions corresponding to the NAMEs are so marked. If no
- arguments are given, or if -p' is given, a list of all readonly names is printed. The -a' option means to treat each NAME as
- an array variable. An argument of --' disables further option processing. NAMEs are marked for automatic export to the environment of subsequently executed commands. If the -f option is given, the NAMEs refer to functions. If no NAMEs are given, or if -p'
- is given, a list of all names that are exported in this shell is
- printed. An argument of -n' says to remove the export property from subsequent NAMEs. An argument of --' disables further option
- processing.
- For each NAME, remove the corresponding variable or function. Given
- the -v', unset will only act on variables. Given the -f' flag,
- unset will only act on functions. With neither flag, unset first
- tries to unset a variable, and if that fails, then tries to unset a
- function. Some variables cannot be unset; also see readonly.
- -a Mark variables which are modified or created for export.
- -b Notify of job termination immediately.
- -e Exit immediately if a command exits with a non-zero status.
- -f Disable file name generation (globbing).
- -h Remember the location of commands as they are looked up.
- -k All assignment arguments are placed in the environment for a
- command, not just those that precede the command name.
- -m Job control is enabled.
- -n Read commands but do not execute them.
- -o option-name
- Set the variable corresponding to option-name:
- allexport same as -a
- braceexpand same as -B
- emacs use an emacs-style line editing interface
- errexit same as -e
- errtrace same as -E
- functrace same as -T
- hashall same as -h
- histexpand same as -H
- history enable command history

- *ignoreeof the shell will not exit upon reading EOF*
- *interactive-comments*
- *allowcomments to appear in interactive commands*
- *keyword same as -k*
- *monitor same as -m*
- *noclobber same as -C*
- *noexec same as -n*
- *noglob same as -f*
- *nolog currently accepted but ignored*
- *notify same as -b*
- *nounset same as -u*
- *onecmd same as -t*
- *physical same as -P*
- *pipefail the return value of a pipeline is the status of*
- *the last command to exit with a non-zero status,*
- *or zero if no command exited with a non-zero status*
- *posix change the behavior of bash where the default*
- *operation differs from the 1003.2 standard to*
- *match the standard*
- *privileged same as -p*
- *verbose same as -v*
- *vi use a vi-style line editing interface*
- *xtrace same as -x*
- *-p Turned on whenever the real and effective user ids do not match.*
- *Disables processing of the $ENV file and importing of shell*
- *functions. Turning this option off causes the effective uid and*
- *gid to be set to the real uid and gid.*
- *-t Exit after reading and executing one command.*
- *-u Treat unset variables as an error when substituting.*
- *-v Print shell input lines as they are read.*
- *-x Print commands and their arguments as they are executed.*
- *-B the shell will perform brace expansion*
- *-C If set, disallowexisting regular files to be overwritten*
- *by redirection of output.*
- *-E If set, the ERR trap is inherited by shell functions.*
- *-H Enable ! style history substitution. This flag is on*
- *by default when the shell is interactive.*
- *-P If set, do not followsymbolic links when executing commands*
- *such as cd which change the current directory.*
- *-T If set, the DEBUG trap is inherited by shell functions.*
-
    - *Assign any remaining arguments to the positional parameters.*
- *The -x and -v options are turned off.*
- *Using + rather than - causes these flags to be turned off. The*
- *flags can also be used upon invocation of the shell. The current*
- *set of flags may be found in $-. The remaining n ARGs are positional*
- *parameters and are assigned, in order, to $1, $2, .. $n. If no*
- *ARGs are given, all shell variables are printed.*
- *Causes a function to exit with the return value specified by N. If N*
- *is omitted, the return status is that of the last command.*
- *One line is read from the standard input, or from file descriptor FD if the*
- *-u option is supplied, and the first word is assigned to the first NAME,*
- *the second word to the second NAME, and so on, with leftover words assigned*

- to the last NAME. Only the characters found in $IFS are recognized as word
- delimiters. If no NAMEs are supplied, the line read is stored in the REPLY
- variable. If the -r option is given, this signifies raw' input, and backslash escaping is disabled. The -d option causes read to continue until the first character of DELIM is read, rather than newline. If the -p option is supplied, the string PROMPT is output without a trailing newline before attempting to read. If -a is supplied, the words read are assigned to sequential indices of ARRAY, starting at zero. If -e is supplied and the shell is interactive, readline is used to obtain the line. If -n is supplied with a non-zero NCHARS argument, read returns after NCHARS characters have been read. The -s option causes input coming from a terminal to not be echoed. The -t option causes read to time out and return failure if a complete line of input is not read within TIMEOUT seconds. If the TMOUT variable is set, its value is the default timeout. The return code is zero, unless end-of-file is encountered, read times out, or an invalid file descriptor is supplied as the argument to -u. Each ARG is an arithmetic expression to be evaluated. Evaluation is done in fixed-width integers with no check for overflow, though division by 0 is trapped and flagged as an error. The following list of operators is grouped into levels of equal-precedence operators. The levels are listed in order of decreasing precedence. id++, id-- variable post-increment, post-decrement ++id, --id variable pre-increment, pre-decrement unary minus, plus logical and bitwise negation exponentiation *, /, % multiplication, division, remainder addition, subtraction <<, >> left and right bitwise shifts <=, >=, <, > comparison ==, != equality, inequality bitwise AND bitwise XOR bitwise OR logical AND logical OR expr ? expr : expr conditional operator =, *=, /=, %=, +=, -=, <<=, >>=, &=, ^=, |= assignment Shell variables are allowed as operands. The name of the variable is replaced by its value (coerced to a fixed-width integer) within an expression. The variable need not have its integer attribute turned on to be used in an expression. Operators are evaluated in order of precedence. Sub-expressions in parentheses are evaluated first and may override the precedence rules above. If the last ARG evaluates to 0, let returns 1; 0 is returned otherwise. Send the processes named by PID (or JOBSPEC) the signal SIGSPEC. If SIGSPEC is not present, then SIGTERM is assumed. An argument of -l'
- lists the signal names; if arguments follow -l' they are assumed to be signal numbers for which names should be listed. Kill is a shell builtin for two reasons: it allows job IDs to be used instead of process IDs, and, if you have reached the limit on processes that you can create, you don't have to start a process to kill another one. By default, removes each JOBSPEC argument from the table of active jobs. If the -h option is given, the job is not removed from the table, but is marked so that SIGHUP is not sent to the job if the shell receives a SIGHUP. The -a option, when JOBSPEC is not supplied, means to remove all jobs from the job table; the -r option means to remove only running jobs. Lists the active jobs. The -l option lists process id's in addition to the normal information; the -p option lists process id's only. If -n is given, only processes that have changed status since the last notification are printed. JOBSPEC restricts output to that job. The -r and -s options restrict output to running and stopped jobs only, respectively. Without options, the status of all active jobs is printed. If -x is given, COMMAND is run after all job specifications that appear in ARGS have been replaced with the process ID of that job's process group leader. Display the history list with line numbers. Lines listed with with a ' have been modified. Argument of N says to list only
- the last N lines. The -c' option causes the history list to be cleared by deleting all of the entries. The -d' option deletes
- the history entry at offset OFFSET. The -w' option writes out the current history to the history file; -r' means to read the file and
- append the contents to the history list instead. -a' means to append history lines from this session to the history file. Argument -n' means to read all history lines not already read
- from the history file and append them to the history list.
- If FILENAME is given, then that is used as the history file else

- if $HISTFILE has a value, that is used, else ~/.bash*history.*
- *If the -s option is supplied, the non-option ARGs are appended to*
- *the history list as a single entry. The -p option means to perform*
- *history expansion on each ARG and display the result, without storing*
- *anything in the history list.*
- *If the $HISTTIMEFORMAT variable is set and not null, its value is used*
- *as a format string for strftime(3) to print the time stamp associated*
- *with each displayed history entry. No time stamps are printed otherwise.*
- *Display helpful information about builtin commands. If PATTERN is*
- *specified, gives detailed help on all commands matching PATTERN,*
- *otherwise a list of the builtins is printed. The -s option*
- *restricts the output for each builtin command matching PATTERN to*
- *a short usage synopsis.*
- *For each NAME, the full pathname of the command is determined and*
- *remembered. If the -p option is supplied, PATHNAME is used as the*
- *full pathname of NAME, and no path search is performed. The -r*
- *option causes the shell to forget all remembered locations. The -d*
- *option causes the shell to forget the remembered location of each NAME.*
- *If the -t option is supplied the full pathname to which each NAME*
- *corresponds is printed. If multiple NAME arguments are supplied with*
- *-t, the NAME is printed before the hashed full pathname. The -l option*
- *causes output to be displayed in a format that may be reused as input.*
- *If no arguments are given, information about remembered commands is displayed.*
- *Place each JOB*SPEC in the background, as if it had been started with
- &'. If JOB_SPEC is not present, the shell's notion of the current job is used. Place JOB_SPEC in the foreground, and make it the current job. If JOB_SPEC is not present, the shell's notion of the current job is used. fc is used to list or edit and re-execute commands from the history list. FIRST and LAST can be numbers specifying the range, or FIRST can be a string, which means the most recent command beginning with that string. -e ENAME selects which editor to use. Default is FCEDIT, then EDITOR, then vi. -l means list lines instead of editing. -n means no line numbers listed. -r means reverse the order of the lines (making it newest listed first). With the fc -s [pat=rep ...] [command]' format, the command is
- re-executed after the substitution OLD=NEW is performed.
- A useful alias to use with this is r='fc -s', so that typing r cc' runs the last command beginning with <code>cc' and typing</code>r' re-executes the last command. Logout of a login shell. Exit the shell with a status of N. If N is omitted, the exit status is that of the last command executed. Exec FILE, replacing this shell with the specified program. If FILE is not specified, the redirections take effect in this shell. If the first argument is -l', then place a dash in the
- zeroth arg passed to FILE, as login does. If the -c' option is supplied, FILE is executed with a null environment. The -a'
- option means to make set argv[0] of the executed process to NAME.
- If the file cannot be executed and the shell is not interactive,
- then the shell exits, unless the shell option execfail' is set. Getopts is used by shell procedures to parse positional parameters. OPTSTRING contains the option letters to be recognized; if a letter is followed by a colon, the option is expected to have an argument, which should be separated from it by white space. Each time it is invoked, getopts will place the next option in the shell variable $name, initializing name if it does not exist, and the index of the next argument to be processed into the shell variable OPTIND. OPTIND is initialized to 1 each time the shell or a shell script is invoked. When an option requires an argument, getopts places that argument into the shell variable OPTARG. getopts reports errors in one of two ways. If the first character of OPTSTRING is a colon, getopts uses silent error reporting. In this mode, no error messages are printed. If an invalid option is seen, getopts places the option

character found into OPTARG. If a required argument is not found, getopts places a ':' into NAME and sets OPTARG to the option character found. If getopts is not in silent mode, and an invalid option is seen, getopts places '?' into NAME and unsets OPTARG. If a required argument is not found, a '?' is placed in NAME, OPTARG is unset, and a diagnostic message is printed. If the shell variable OPTERR has the value 0, getopts disables the printing of error messages, even if the first character of OPTSTRING is not a colon. OPTERR has the value 1 by default. Getopts normally parses the positional parameters ($0 - $9), but if more arguments are given, they are parsed instead. Read ARGs as input to the shell and execute the resulting command(s). Enable and disable builtin shell commands. This allows you to use a disk command which has the same name as a shell builtin without specifying a full pathname. If -n is used, the NAMEs become disabled; otherwise NAMEs are enabled. For example, to use the test' found in $PATH instead of the shell builtin

- version, type enable -n test'. On systems supporting dynamic loading, the -f option may be used to load new builtins from the shared object FILENAME. The -d option will delete a builtin previously loaded with -f. If no non-option names are given, or the -p option is supplied, a list of builtins is printed. The -a option means to print every builtin with an indication of whether or not it is enabled. The -s option restricts the output to the POSIX.2 special' builtins. The -n option displays a list of all disabled builtins.
- Output the ARGs. If -n is specified, the trailing newline is
- suppressed. If the -e option is given, interpretation of the
- following backslash-escaped characters is turned on:
- alert (bell)
- backspace
- suppress trailing newline
- escape character
- form feed
- new line
- carriage return
- horizontal tab
- vertical tab
- backslash
- the character whose ASCII code is NNN (octal). NNN can be
- 0 to 3 octal digits
- You can explicitly turn off the interpretation of the above characters
- with the -E option.
- Create a local variable called NAME, and give it VALUE. LOCAL
- can only be used within a function; it makes the variable NAME
- have a visible scope restricted to that function and its children.
- Obsolete. See declare'. Declare variables and/or give them attributes. If no NAMEs are given, then display the values of variables instead. The -p option will display the attributes and values of each NAME. The flags are: -a to make NAMEs arrays (if supported) -f to select from among function names only -F to display function names (and line number and source file name if debugging) without definitions -i to make NAMEs have the integer' attribute
- -r
- to make NAMEs readonly
- -t
- to make NAMEs have the trace' attribute -x to make NAMEs export Variables with the integer attribute have arithmetic evaluation (see let') done when the variable is assigned to.
- When displaying values of variables, -f displays a function's name
- and definition. The -F option restricts the display to function
- name only.
- Using +' instead of -' turns off the given attribute instead. When

- used in a function, makes NAMEs local, as with the local' command. Runs COMMAND with ARGS ignoring shell functions. If you have a shell function called <code>ls', and you wish to call the command</code>ls', you can say "command ls". If the -p option is given, a default value is used for PATH that is guaranteed to find all of the standard utilities. If the -V or -v option is given, a string is printed describing COMMAND. The -V option produces a more verbose description. Return an unsuccessful result. Return a successful result. No effect; the command does nothing. A zero exit code is returned. Print the current working directory. With the -P option, pwd prints the physical directory, without any symbolic links; the -L option makes pwd follow symbolic links. Change the current directory to DIR. The variable $HOME is the default DIR. The variable CDPATH defines the search path for the directory containing DIR. Alternative directory names in CDPATH are separated by a colon (:). A null directory name is the same as the current directory, i.e. .'. If DIR begins with a slash (/),
- then CDPATH is not used. If the directory is not found, and the
- shell option cdable_vars' is set, then try the word as a variable name. If that variable has a value, then cd to the value of that variable. The -P option says to use the physical directory structure instead of following symbolic links; the -L option forces symbolic links to be followed. Returns the context of the current subroutine call. Without EXPR, returns "$line $filename". With EXPR, returns "$line $subroutine $filename"; this extra information can be used to provide a stack trace. The value of EXPR indicates how many call frames to go back before the current one; the top frame is frame 0. Run a shell builtin. This is useful when you wish to rename a shell builtin to be a function, but need the functionality of the builtin within the function itself. Resume the next iteration of the enclosing FOR, WHILE or UNTIL loop. If N is specified, resume at the N-th enclosing loop. Exit from within a FOR, WHILE or UNTIL loop. If N is specified, break N levels. Bind a key sequence to a Readline function or a macro, or set a Readline variable. The non-option argument syntax is equivalent to that found in ~/.inputrc, but must be passed as a single argument: bind '"\C-x\C-r": re-read-init-file'. bind accepts the following options: -m keymap Use keymap' as the keymap for the duration of this
- command. Acceptable keymap names are emacs,
- emacs-standard, emacs-meta, emacs-ctlx, vi, vi-move,
- vi-command, and vi-insert.
- -l List names of functions.
- -P List function names and bindings.
- -p List functions and bindings in a form that can be
- reused as input.
- -r keyseq Remove the binding for KEYSEQ.
- -x keyseq:shell-command
- Cause SHELL-COMMAND to be executed when
- KEYSEQ is entered.
- -f filename Read key bindings from FILENAME.
- -q function-name Query about which keys invoke the named function.
- -u function-name Unbind all keys which are bound to the named function.
- -V List variable names and values
- -v List variable names and values in a form that can
- be reused as input.
- -S List key sequences that invoke macros and their values
- -s List key sequences that invoke macros and their values
- in a form that can be reused as input.
- Remove NAMEs from the list of defined aliases. If the -a option is given,
- then remove all alias definitions.
- alias' with no arguments or with the -p option prints the list of aliases in the form alias NAME=VALUE on standard output. Otherwise, an alias is defined for each NAME whose VALUE is given. A trailing space in VALUE causes the next word to be checked for alias substitution when

```
the alias is expanded. Alias returns</li> <li>true unless a NAME is given for which no alias has been
defined.</li> <li>alias [-p] [name[=value] ... ]</li> <li>unalias</li> <li>unalias [-a] name [name ...]</li>
<li>bind [-lpvsPVS] [-m keymap] [-f filename] [-q name] [-u name] [-r keyseq] [-x keyseq:shell-command]
</li> <li>break [n]</li> <li>continue</li> <li>continue [n]</li> <li>builtin</li> <li>builtin [shell-builtin
[arg ...]]</li> <li>caller</li> <li>caller [EXPR]</li> <li>cd [-L|-P] [dir]</li> <li>pwd [-LP]</li>
<li>command</li> <li>command [-pVv] command [arg ...]</li> <li>declare</li> <li>declare [-afFirtx] [-p]
[name[=value] ...]</li> <li>typeset</li> <li>typeset [-afFirtx] [-p] name[=value] ...</li> <li>local
name[=value] ...</li> <li>echo [-neE] [arg ...]</li> <li>enable</li> <li>enable [-pnds] [-a] [-f filename]
[name ...]</li> <li>eval [arg ...]</li> <li>getopts</li> <li>getopts optstring name [arg]</li> <li>exec [-
cl] [-a name] file [redirection ...]</li> <li>exit [n]</li> <li>fc [-e ename] [-nlr] [first] [last] or fc -s
[pat=rep] [cmd]</li> <li>fg [job_spec]</li> <li>bg [job_spec ...]</li> <li>hash [-lr] [-p pathname] [-dt]
[name ...]</li> <li>help [-s] [pattern ...]</li> <li>history [-c] [-d offset] [n] or history -awrn
[filename] or history -ps arg [arg...]</li> <li>jobs [-lnprs] [jobspec ...] or jobs -x command [args]</li>
<li>disown</li> <li>disown [-h] [-ar] [jobspec ...]</li> <li>kill [-s sigspec | -n signum | -sigspec] pid |
jobspec ... or kill -l [sigspec]</li> <li>let arg [arg ...]</li> <li>read [-ers] [-u fd] [-t timeout] [-p
prompt] [-a array] [-n nchars] [-d delim] [name ...]</li> <li>return</li> <li>return [n]</li> <li>set [--
abefhkmnptuvxBCHP] [-o option] [arg ...]</li> <li>unset [-f] [-v] [name ...]</li> <li>export</li> <li>export
[-nf] [name[=value] ...] or export -p</li> <li>readonly</li> <li>readonly [-af] [name[=value] ...] or
readonly -p</li> <li>shift [n]</li> <li>source filename [arguments]</li> <li>. filename [arguments]</li>
<li>suspend</li> <li>suspend [-f]</li> <li>test [expr]</li> <li>[ arg... ]</li> <li>trap [-lp] [arg
signal_spec ...]</li> <li>type [-afptP] name [name ...]</li> <li>ulimit</li> <li>ulimit [-SHacdfilmnpqstuvx]
[limit]</li> <li>umask [-p] [-S] [mode]</li> <li>wait [n]</li> <li>for NAME [in WORDS ... ;] do COMMANDS;
done</li> <li>for (( exp1; exp2; exp3 )); do COMMANDS; done</li> <li>select NAME [in WORDS ... ;] do
COMMANDS; done</li> <li>time [-p] PIPELINE</li> <li>case WORD in [PATTERN [| PATTERN]...) COMMANDS ;;]...
esac</li> <li>if COMMANDS; then COMMANDS; [ elif COMMANDS; then COMMANDS; ]... [ else COMMANDS; ] fi</li>
<li>while COMMANDS; do COMMANDS; done</li> <li>until COMMANDS; do COMMANDS; done</li> <li>function NAME {
COMMANDS ; } or NAME () { COMMANDS ; }</li> <li>{ ... }</li> <li>{ COMMANDS ; }</li> <li>JOB_SPEC [&amp;]
</li> <li>(( ... ))</li> <li>(( expression ))</li> <li>[[ ... ]]</li> <li>[[ expression ]]</li>
<li>variables</li> <li>variables - Some variable names and meanings</li> <li>pushd [dir | +N | -N] [-n]</li>
<li>popd [+N | -N] [-n]</li> <li>dirs [-clpv] [+N] [-N]</li> <li>shopt [-pqsu] [-o long-option] optname
[optname...]</li> <li>printf</li> <li>printf [-v var] format [arguments]</li> <li>complete</li> <li>complete
[-abcdefgjksuv] [-pr] [-o option] [-A action] [-G globpat] [-W wordlist] [-P prefix] [-S su</li>
<li>compgen</li> <li>compgen [-abcdefgjksuv] [-o option] [-A action] [-G globpat] [-W wordlist] [-P prefix]
[-S suffix] [</li> <li>alias </li> <li>
```
%s': invalid alias name

- bind*builtin*
- *lvpVPsSf:q:u:m:r:x:*
- *%s': invalid keymap name %s: cannot read: %s %s': unknown function name*
- *%s is not bound to any keys.*
- *%s can be invoked via*
- *%s': cannot unbind only meaningful in a <code>for',</code>while', or until' loop*
- *loop count*
- *%s %s %s*
- *OLDPWD=*
- *HOME not set*
- *OLDPWD not set*
- *CDPATH*
- *command*builtin
- /bin:/usr/bin:/sbin:/usr/sbin:/etc:/usr/etc
- line %d:
- %s: usage:
- %s: numeric argument required
- %s: not found
- %s: invalid option name

- %s: invalid number
- %s: invalid signal specification
- %s': not a pid or valid job spec %s: %s out of range argument %s out of range %s: no such job %s: no job control no job control %s: restricted %s: not a shell builtin write error: %s %s: error retrieving current directory: %s: %s %s: ambiguous job spec SIGJUNK Unknown %2d) %s +afiprtxF cannot use -f' to make functions
- %s %d %s
- %s: cannot destroy array variables in this way
- can only be used in a function
- adnpsf:
- enable %s
- enable -n %s
- cannot open shared object %s: %s
- *struct*
- *cannot find %s in shared object %s: %s*
- *%s: not dynamically loaded*
- *%s: cannot delete: %s*
- *%s: not a regular file*
- *%s: file is too large*
- evalfile
- source
- parse*and*execute*top*
- *pe*dispose
- parse*and*execute
- %s: cannot execute: %s
- ~/.bash*logout*
- *There are stopped jobs.*
- *not login shell: use* exit' logout :e:lnrs no command found history specification bash-fc %s: cannot open temp file: %s ${FCEDIT:-${EDITOR:-ed}} ${FCEDIT:-${EDITOR:-vi}} fc builtin job %d started without job control builtin hash -p %s %s hashing disabled dlp:rt command %s: hash table empty These shell commands are defined internally. Type *help' to see this list.*
- *Type* help name' to find out more about the function *name'.*
- *Use* info bash' to find out more about the shell in general. Use <code>man -k' or</code>info' to find out more about commands not in this list. A star (*) next to a name means that the command is disabled. Shell commands matching keywords
- *Shell commands matching keyword* %s: %s %s: cannot open: %s no help topics match <code>%s'. Try</code>help help' or <code>man -k %s' or</code>info %s'. acd:npsrw cannot use more than one of -anrw %s: history expansion failed history position %5d%c %s%s no other options allowed with *-x'*
- *lpnxrs*
- *jobs*builtin
- current
- Unknown error
- (%ld) - %s
- %s: arguments must be process or job IDs
- directory stack empty
- directory stack index
- dirstack
-
- %2d %s
- %2d %s

- no other directory
- %s: invalid timeout specification
- %s: invalid file descriptor specification
- %d: invalid file descriptor: %s
- ersa:d:n:p:t:u:
- read*builtin*
- *read error: %d: %s*
- *can only* return' from a function or sourced script allexport braceexpand errexit errtrace functrace hashall histexpand history interactive-comments keyword monitor noclobber noexec noglob notify nounset onecmd physical pipefail privileged xtrace SHELLOPTS cannot simultaneously unset a function and a variable %s: cannot unset %s: cannot unset: readonly %s variable %s: not an array variable declare -%s %s -%s %s: not a function shift count filename argument required %s: file not found cannot suspend cannot suspend a login shell unknown trap -- %s %d trap -- %s %s %s is aliased to *%s'*
- *alias %s=%s*
- *%s is a shell keyword*
- *%s is a function*
- *%s is a shell builtin*
- *%s is %s*
- *%s is hashed (%s)*
- *core file size*
- *blocks*
- *data seg size*
- *kbytes*
- *file size*
- *max locked memory*
- *max memory size*
- *open files*
- *pipe size*
- *512 bytes*
- *stack size*
- *cpu time*
- *seconds*
- *max user processes*
- *virtual memory*
- *(%s, -%c)*
- *(-%c)*
- *%-20s %16s*
- *unlimited*
- *%s: cannot get limit: %s*
- %c': bad command %s: cannot modify limit: %s u=%s,g=%s,o=%s *%c': invalid symbolic mode operator*
- %c': invalid symbolic mode character octal number umask%s OPTARG cdable_vars cdspell checkhash checkwinsize cmdhist dotglob execfail expand_aliases extdebug extglob extquote failglob force_fignore gnu_errfmt histappend histreedit histverify hostcomplete huponexit interactive_comments lithist login_shell mailwarn noemptycmd_completion nocaseglob nocasematch nullglob progcomp promptvars restricted_shell shift_verbose sourcepath

xpg_echo %s: invalid shell option name shopt %s %s cannot set and unset shell options simultaneously set %co %s missing hex digit for \x warning: %s: %s #'-+ 0 hjlLtz *%s': missing format character*

- %c': invalid format character %s: illegal option -- %c %s: option requires an argument -- %c arrayvar binding directory disabled enabled helptopic hostname running service setopt signal stopped bashdefault default dirnames filenames nospace plusdirs %s: invalid action name abcdefgjko:prsuvA:G:W:P:S:X:F:C: complete -o %s -A %s %s %s warning: -F option may not work as you expect warning: -C option may not work as you expect %s: no completion specification BASH_REMATCH /dev/std %s: bad network path specification socket connect %ldm%d.%03ds sh_makepath sh_realpath TMPDIR /var/tmp/ /var/tmp /usr/tmp %s/%s-%lu %s/cur %s/tmp %s/new invalid base 0123456789ABCDEF 0123456789abcdef hl^$0ftFT;,%wbeWBE| _*\AaIiCcDdPpYyRrSsXx~ l|h^0%bB l|h^0bB l|hwW^0bB accept-line arrow-key-prefix backward-byte backward-char backward-delete-char backward-kill-line backward-kill-word backward-word beginning-of-history beginning-of-line call-last-kbd-macro capitalize-word character-search character-search-backward clear-screen copy-backward-word copy-forward-word copy-region-as-kill delete-char delete-char-or-list delete-horizontal-space digit-argument do-lowercase-version downcase-word dump-functions dump-macros dump-variables emacs-editing-mode end-kbd-macro end-of-history end-of-line exchange-point-and-mark forward-backward-delete-char forward-byte forward-char forward-search-history forward-word history-search-backward history-search-forward insert-comment insert-completions kill-whole-line kill-line kill-region kill-word menu-complete next-history non-incremental-forward-search-history non-incremental-reverse-search-history non-incremental-forward-search-history-again non-incremental-reverse-search-history-again overwrite-mode possible-completions previous-history quoted-insert re-read-init-file redraw-current-line reverse-search-history revert-line self-insert set-mark start-kbd-macro tab-insert tilde-expand transpose-chars transpose-words tty-status universal-argument unix-filename-rubout unix-line-discard unix-word-rubout upcase-word yank-last-arg yank-nth-arg yank-pop vi-append-eol vi-append-mode vi-arg-digit vi-back-to-indent vi-bWord vi-bword vi-change-case vi-change-char vi-change-to vi-char-search vi-column vi-complete vi-delete vi-delete-to vi-eWord vi-editing-mode vi-end-word vi-eof-maybe vi-eword vi-fWord vi-fetch-history vi-first-print vi-fword vi-goto-mark vi-insert-beg vi-insertion-mode vi-match vi-movement-mode vi-next-word vi-overstrike vi-overstrike-delete vi-prev-word vi-put vi-redo vi-replace vi-rubout vi-search vi-search-again vi-set-mark vi-subst vi-tilde-expand vi-yank-arg vi-yank-to "\' @$><=;|&{(
- *--More--*
- *Display all %d possibilities? (y or n)*
- *readline: bad value %d for whattodo in rl*complete
- Control-
- bell-style
- completion-query-items
- editing-mode
- isearch-terminators
- keymap

- bind-tty-special-chars
- blink-matching-paren
- byte-oriented
- convert-meta
- disable-completion
- enable-keypad
- expand-tilde
- history-preserve-point
- horizontal-scroll-mode
- input-meta
- mark-directories
- mark-modified-lines
- mark-symlinked-directories
- match-hidden-files
- meta-flag
- output-meta
- page-completions
- prefer-visible-bell
- print-completions-horizontally
- show-all-if-ambiguous
- show-all-if-unmodified
- visible-stats
- emacs-standard
- emacs-meta
- emacs-ctlx
- vi-move
- vi-command
- vi-insert
- include
- Escape
- Newline
- Return
- Rubout
- readline: %s: line %d: %s
- readline: %s
- $else found without matching $if
- $endif without matching $if
- audible
- visible
- # %s (not bound)
- "%s": %s
- %s is not bound to any keys
- %s can be found on
- "%s"%s
- set %s %s
- %s is set to `%s'</li> <li>"%s%s": "%s"</li> <li>%s%s outputs %s</li> <li>unknown parser directive</li> <li>no closing`"' in key binding
- prefix-meta
- INPUTRC
- ~/.inputrc
- /etc/inputrc
- reverse-
- i-search)`</li> <li>readline: debug: insert<em>some</em>chars: count (%d) != col (%d)</li> <li>/-_=~.#$</li>`

```
<li>readline: readline<em>callback</em>read_char() called with no handler!</li> <li>(arg: %d) </li>
<li>;&amp;()|&lt;&gt;</li> <li>no previous substitution</li> <li>unknown expansion error</li> <li>event not
found</li> <li>bad word specifier</li> <li>substitution failed</li> <li>unrecognized history modifier</li>
<li>&lt;&gt;;&amp;|$</li> <li>.history</li> <li>xdigit</li> <li>backspace</li> <li>vertical-tab</li>
<li>form-feed</li> <li>carriage-return</li> <li>exclamation-mark</li> <li>quotation-mark</li> <li>number-
sign</li> <li>dollar-sign</li> <li>percent-sign</li> <li>ampersand</li> <li>apostrophe</li> <li>left-
parenthesis</li> <li>right-parenthesis</li> <li>asterisk</li> <li>plus-sign</li> <li>hyphen</li> <li>hyphen-
minus</li> <li>period</li> <li>full-stop</li> <li>solidus</li> <li>semicolon</li> <li>less-than-sign</li>
<li>equals-sign</li> <li>greater-than-sign</li> <li>question-mark</li> <li>commercial-at</li> <li>left-
square-bracket</li> <li>backslash</li> <li>reverse-solidus</li> <li>right-square-bracket</li>
<li>circumflex</li> <li>circumflex-accent</li> <li>underscore</li> <li>grave-accent</li> <li>left-brace</li>
<li>left-curly-bracket</li> <li>vertical-line</li> <li>right-brace</li> <li>right-curly-bracket</li>
<li>/locale.alias</li> <li>/usr/share/locale</li> <li>messages</li> <li>LC_XXX</li> <li>LC_MONETARY</li>
<li>LANGUAGE</li> <li>GETTEXT<em>LOG</em>UNTRANSLATED</li> <li>plural=</li> <li>nplurals=</li> <li>domain
</li> <li>msgid </li> <li>msgid_plural </li> <li>msgstr[0] ""</li> <li>msgstr ""</li> <li>!"#$&amp;'(),-
/02479:&lt;=&gt;?@ABCEFGHIVWXYZ,OPU</li> <li>4 !"#$&amp;()/0</li> <li>:9;,-S</li> <li>,QRXU+U</li>
<li>359UUWWUYP</li> <li>2PP5>DUWW*+Y</li> <li>JUUUUUKLMU</li> <li>TU3PPOO</li> <li>3UPUUU5</li>
<li>67777889999999999999999999999999999:::;;&lt;&lt;====&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;&gt;????????
@@@@AAAAAABBBCCCDDEFFFGHIJJJKKLLLL</li> <li>-0:41/65789</li> <li>%+(.uv</li> <li>zzzzzz</li> <li>EFzzzz</li>
<li>XopqrsOe
```

- zHQWzzzzfN
- zzCDIJ
- zhabzz
- zzz^cd
- !"#$%&'()-
- *mhexecute*header
- *_DefaultRuneLocale*
- divdi3
- *error*
- maskrune
- *mbcurmax*
- moddi3
- *stderrp*
- stdinp
- *stdoutp*
- tolower
- *toupper*
- udivdi3
- *umoddi3*
- abort
- *access*
- alarm
- *asprintf$LDBL128*
- bcopy
- *bsearch*
- chdir
- *clearerr*
- close$UNIX2003
- *closedir$UNIX2003*
- *compatmode*
- confstr$UNIX2003
- *connect$UNIX2003*
- dlclose

- *dlerror*
- dlopen
- *dlsym*
- endgrent
- *endpwent*
- endservent
- *execve*
- fclose
- *fcntl$UNIX2003*
- fdopen
- *ferror*
- fflush
- *fgets*
- fileno
- *fopen*
- fprintf$LDBL128
- *fputc*
- fputs$UNIX2003
- *freeaddrinfo*
- fstat
- *fwrite$UNIX2003*
- *gaistrerror*
- getaddrinfo
- *getcwd*
- getdtablesize
- *getegid*
- geteuid
- *getgid*
- getgrent
- *getgroups*
- gethostname
- *getpeername$UNIX2003*
- getpgrp
- *getpid*
- getppid
- *getpwent*
- getpwnam
- *getpwuid*
- getrlimit$UNIX2003
- *getrusage*
- getservent
- *gettimeofday*
- getuid
- *ioctl*
- isatty
- *kill$UNIX2003*
- killpg$UNIX2003
- *localtime*
- lseek
- *lstat*
- malloc
- *mblen*

- mbrlen
- *mbrtowc*
- mbsinit
- *mbsrtowcs*
- mbstowcs
- *mbtowc*
- memcmp
- *memcpy*
- memmove
- *memset*
- mmap$UNIX2003
- *munmap$UNIX2003*
- open$UNIX2003
- *opendir$UNIX2003*
- pathconf
- *printf$LDBL128*
- putchar
- *qsort*
- raise
- *read$UNIX2003*
- readdir
- *readlink*
- realloc
- *regcomp$UNIX2003*
- regexec
- *regfree*
- select$UNIX2003
- *setgid*
- setgrent
- *setlocale*
- setpgid
- *setpwent*
- setregid$UNIX2003
- *setreuid$UNIX2003*
- setrlimit$UNIX2003
- *setservent*
- setuid
- *setvbuf*
- sigaction
- *sigaddset*
- sigemptyset
- *siglongjmp*
- sigprocmask
- *sigsetjmp*
- sleep$UNIX2003
- *snprintf$LDBL128*
- socket
- *sprintf$LDBL128*
- stpcpy
- *strcasecmp*
- strcat
- *strchr*

- strcmp
- *strcoll*
- strcpy
- *strdup*
- strerror$UNIX2003
- *strftime$UNIX2003*
- strlen
- *strncasecmp*
- strncmp
- *strncpy*
- strpbrk
- *strrchr*
- strsignal
- *strstr*
- strtod$UNIX2003
- *strtoimax*
- strtoul
- *strtoumax*
- sysconf
- *tcgetattr*
- tcgetpgrp
- *tcsetattr*
- tcsetpgrp
- *tgetent*
- tgetflag
- *tgetnum*
- tgetstr
- *tgoto*
- tputs
- *ttyname*
- tzset
- *umask*
- unlink
- *vfprintf$LDBL128*
- vsnprintf$LDBL128
- *waitpid$UNIX2003*
- wcrtomb
- *wcschr*
- wcscmp
- *wcscoll*
- wcscpy
- *wcslen*
- wcsncmp
- *wcsrtombs*
- wctob
- *wctype*
- wcwidth
- *write$UNIX2003*
- *com.apple.bash*
- *my*01R
- gfONe
- *44pCRJ*

- *L:GIRS*
- U70t6
- com.apple.translate
- Apple Inc.1&0$
- Apple Certification Authority1
- Apple Root CA0
- 070214211919Z
- 150214211919Z0
- Apple Inc.1&0$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority0*
- *drm|iGa*
- *%http://www.apple.com/appleca/root.crl0*
- *QKK:IaN*
- *Apple Inc.1&0$*
- *Apple Certification Authority1*
- *Apple Root CA0*
- *060425214036Z*
- *350209214036Z0b1*
- *Apple Inc.1&0$*
- *Apple Certification Authority1*
- *Apple Root CA0*
- *https://www.apple.com/appleca/0*
- *Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a*
- *Apple Inc.1&0$*
- *Apple Certification Authority1301*
- Apple Code Signing Certification Authority0
- 070223220256Z
- 150114220256Z0V1
- Apple Inc.1
- Apple Software1
- Software Signing0
- EA3%|k=
- http://www.apple.com/appleca/0
- Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a
- ,http://www.apple.com/appleca/codesigning.crl0
- Apple Inc.1&0$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority
- 0bX{7<

UTF-8 Strings

No detected.

UTF-16LE Strings

- XdiouxX
- !c"#$%&'()
- 56tufp7/
-]^T9Uwy:~sz
- mnogh_
- 244.,,
- &'()55

- " !"#$3&'()
- 4 !"#$
- !c"#$%&'()
- 56tufp7/
- L<=@b1
-]^T9Uwy:~sz
- mnogh_
- 244.,,
- &'()55
- " !"#$3&'()

Extracted URLs

- http://www.apple.com/appleca/root.crl0
- https://www.apple.com/appleca/0
- http://www.apple.com/appleca/0
- http://www.apple.com/appleca/codesigning.crl0
- http://www.apple.com/appleca/root.crl0
- https://www.apple.com/appleca/0
- http://www.apple.com/appleca/0
- http://www.apple.com/appleca/codesigning.crl0

Base64 Strings

No valid Base64-encoded strings found.

File Entropy Analysis

- **Entropy**: `6.41`
- **Status**: This file is likely packed or encrypted.

VirusTotal Analysis

Scan Date: 2021-08-08 19:33:48

Scan Results

- Malicious: 0
- Suspicious: 0
- Undetected: 60
- Harmless: 0
- Timeout: 0
- Confirmed-timeout: 0
- Failure: 0
- Type-unsupported: 14

Community Votes

- Harmless: 0
- Malicious: 0

Detailed Engine Results

Engine Name	Detection Category	Detection
No results	No results	No results